COUNTRY PROFILES

Nuclear Power in Russia

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- Russia is moving steadily forward with plans for an expanded role of nuclear energy, including development of new reactor technology.
- It is committed to closing the fuel cycle, and sees fast reactors as key to this.
- Exports of nuclear goods and services are a major Russian policy and economic objective. Russia is currently involved in the construction of about 20 reactors abroad.
- Russia is a world leader in fast neutron reactor technology and is consolidating this through its *Proryv* ('Breakthrough') project.

34

OPERABLE REACTORS

27,969 MWe

7

REACTORS UNDER CONSTRUCTION

5,002 MWe

14

REACTORS SHUTDOWN

4,929 MWe

Operable nuclear power capacity

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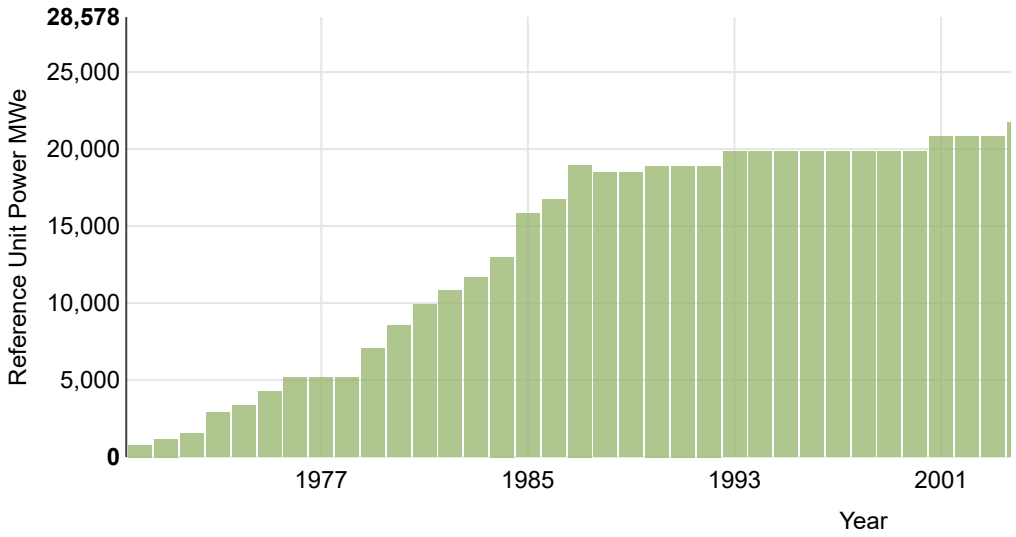


Electricity sector

Total generation (in 2023): 1162 TWh

Generation mix: natural gas 511 TWh (44%); nuclear 217 TWh (19%); coal 211 TWh (18%); hydro 202 TWh (17%); oil 9.3 TWh (1%); wind 4.8 TWh; biofuels & waste 3.9 TWh; solar 2.7 TWh.

Import/export balance: 16.53 TWh net export (1.6 TWh imports; 18.1 TWh exports)

Total consumption: 837 TWh

Per capita consumption: c. 5800 kWh in 2023

Source: International Energy Agency. Data for year 2023.

Russia is one of the few countries without a populist energy policy favouring wind and solar generation; the priority is unashamedly nuclear.

Industry development

Russia's first nuclear power plant, and the first in the world to produce electricity in 1954, was the 5 MWe Obninsk reactor. Russia's first two commercial-scale nuclear power plants started up in 1963-64, then in 1971-73 the first of today's production models were commissioned. By the mid-1980s Russia had 25 power reactors in operation, but the nuclear industry was beset by problems. The Chernobyl accident led to a resolution of these, as outlined in the Appendix of the information page on *Russia's Nuclear Fuel Cycle*.

Between the 1986 Chernobyl accident and the mid-1990s, only one nuclear power station was commissioned in Russia, the four-unit Balakovo, with unit 3 being added to Smolensk. Economic reforms following the collapse of the Soviet Union meant an acute shortage of funds for nuclear developments, and a number of projects were stalled. But by the late 1990s exports of reactors to Iran, China and India were negotiated and Russia's stalled domestic construction programme was revived as far as funds allowed.

Around 2000, nuclear construction revived and Rostov 1 (also known as Volgodonsk 1), the first of the delayed units, started up in 2001, joining 21 GWe already on the grid. It was followed by Kalinin 3 in 2004, Rostov 2 in 2010 and Kalinin 4 in 2011.

By 2006 the government's resolve to develop nuclear power had firmed and there were projections of adding 2-3 GWe per year to 2030 in Russia as well as exporting plants to meet world demand for some 300 GWe of new nuclear capacity in that timeframe. Early in 2016 Rosatom said that Russia's GDP gained three roubles for every one rouble invested in building nuclear power plants domestically, as well as enhanced "socio-economic development of the country as a whole."

In 2017 the CEO of Rosatom said that the government would end state support for the construction of new nuclear units in 2020, and so Rosatom must learn to earn money on its own, primarily via commercial nuclear energy projects in the international market. He said that Rosatom had come from being a consortium of unprofitable, separately-run businesses a decade ago to a vertically-integrated state corporation with improved strategies and financial performance, thanks in part to a "large-scale" programme of state funding. "In this situation ... we must learn how to earn money independently," especially in the world market. "Optimisation of the management system should become the main theme of 2017."

Rosatom's long-term strategy up to 2050 involves moving to inherently safe nuclear plants using fast reactors with a closed fuel cycle, especially under the *Proryv* ('Breakthrough') project. It envisages nuclear providing 45-50% of electricity at that time, with the share rising to 70-80% by the end of the century. The ultimate aim of the closed fuel cycle is to eliminate the production of radioactive waste from power generation.

Apart from adding capacity, utilization of existing plants has improved markedly since 2000. In the 1990s capacity factors

averaged around 60%, but they have steadily improved since and are now above 80%.

See also subsections: Transition to Fast Reactors, and Fast reactors in the *Reactor Technology* section below.

Nuclear power plants

Russia has 34 operable reactors. Rosenergoatom is the only Russian utility operating nuclear power plants. Its nuclear plants have the status of branches. It was established in 1992 and was reconstituted as a utility in 2001, as a division of Rosatom.

Operable reactors in Russia

Reactor name	Model	Reactor Type	Net capacity (MWe)	Construction start	Grid connection
Akademik Lomonosov 1	KLT-40S 'Floating'	PWR	32	2007-04-15	2019-12-19
Akademik Lomonosov 2	KLT-40S 'Floating'	PWR	32	2007-04-15	2019-12-19
Balakovo 1	VVER V-320	PWR	950	1980-12-01	1985-12-28
Balakovo 2	VVER V-320	PWR	950	1981-08-01	1987-10-08
Balakovo 3	VVER V-320	PWR	950	1982-11-01	1988-12-25
Balakovo 4	VVER V-320	PWR	950	1984-04-01	1993-04-11
Beloyarsk 3	BN-600	FBR	560	1969-01-01	1980-04-08
Beloyarsk 4	BN-800	FBR	820	2006-07-18	2015-12-10
Kalinin 1	VVER V-338	PWR	950	1977-02-01	1984-05-09
Kalinin 2	VVER V-338	PWR	950	1982-02-01	1986-12-03
Kalinin 3	VVER V-320	PWR	950	1985-10-01	2004-12-16
Kalinin 4	VVER V-320	PWR	950	1986-08-01	2011-11-24

Reactor name	Model	Reactor Type	Net capacity (MWe)	Construction start	Grid connection
Kola 1	VVER V-230	PWR	411	1970-05-01	1973-06-29
Kola 2	VVER V-230	PWR	411	1970-05-01	1974-12-09
Kola 3	VVER V-213	PWR	411	1977-04-01	1981-03-24
Kola 4	VVER V-213	PWR	411	1976-08-01	1984-10-11
Kursk 2-1	VVER-TOI	PWR	1101	2018-04-28	2025-12-31
Kursk 3	RBMK-1000	LWGR	925	1978-04-01	1983-10-17
Kursk 4	RBMK-1000	LWGR	925	1981-05-01	1985-12-02
Leningrad 2 1	VVER V-491	PWR	1101	2008-10-25	2018-03-09
Leningrad 2 2	VVER V-491	PWR	1101	2010-04-15	2020-10-22
Leningrad 3	RBMK-1000	LWGR	925	1973-12-01	1979-12-07
Leningrad 4	RBMK-1000	LWGR	925	1975-02-01	1981-02-09
Novovoronezh 2 1	VVER V-392M	PWR	1100	2008-06-24	2016-08-05
Novovoronezh 2 2	VVER V-392M	PWR	1101	2009-07-12	2019-05-01
Novovoronezh 4	VVER V-179	PWR	385	1967-07-01	1972-12-28
Novovoronezh 5	VVER V-187	PWR	950	1974-03-01	1980-05-31
Rostov 1	VVER V-320	PWR	989	1981-09-01	2001-03-30
Rostov 2	VVER V-320	PWR	950	1983-05-01	2010-03-18
Rostov 3	VVER V-320	PWR	950	2009-09-15	2014-12-27

Reactor name	Model	Reactor Type	Net capacity (MWe)	Construction start	Grid connection
Rostov 4	VVER V-320	PWR	979	2010-06-16	2018-02-02
Smolensk 1	RBMK-1000	LWGR	925	1975-10-01	1982-12-09
Smolensk 2	RBMK-1000	LWGR	925	1976-06-01	1985-05-31
Smolensk 3	RBMK-1000	LWGR	925	1984-05-01	1990-01-17

Source: International Atomic Energy Agency and World Nuclear Association

V-320 is the base model of what is generically VVER-1000; V-230 and V-213 are generically VVER-440; V-179 & V-187 are prototypes. Rostov was formerly sometimes known as Volgodonsk. Many reactors have been uprated but current net capacities are mostly unknown.

* Novovoronezh II-1&2 are sometimes referred to as Novovoronezh 6&7.

Lifetime extension, uprates and completing construction

Most reactors are being licensed for operating lifetime extension. Over half of Russia's nuclear generation in 2023 came from units which had been upgraded for long-term operation and were operating beyond their initial design lifetimes (around 30 years).

Generally, Russian reactors were originally licensed for 30 years from first power. Since 2000, licence extensions have been issued for: Beloyarsk 3, Novovoronezh 3-5, Kola 1-4, Kalinin 1&2, Balakovo 1-4, Rostov 1, Kursk 1-4, Leningrad 1-4, Smolensk 1-3, and Bilibino 1-4. Novovoronezh 4&5, Kola 1&2, Rostov 1, Bilibino 2-4, and Leningrad 3&4 have been granted second licence extensions.

Generally the **VVER-440** units have got 15-year operating lifetime extensions. (Kola 1&2 VVER-440 units are V-230 models which the EU has paid to shut down early in countries outside Russia. Novovoronezh 4, a V-179, is a predecessor to these.) The reactor pressure vessels of some older reactors have undergone thermal annealing to reduce acquired brittleness. Kola 1&2 were upgraded for operating lifetime extension to 60 years – 2033 & 2034. Kola 3&4 licence extensions to 2027 and 2029 (45 years) have been confirmed after upgrading work. Novovoronezh 4 is now licensed to operate until 2032, and unit 5 to 2035.

Most **VVER-1000** units are expected to have 30-year operating licence extensions. In 2015 Balakovo 1 was upgraded to extend its operating lifetime to 60 years, followed by the same for unit 2 in 2017, and unit 3 in 2019. Unit 1 was the first large VVER reactor to undergo thermal annealing of the pressure vessel.

RBMK reactors have been uprated and granted 15-year lifetime extensions. Following significant design modifications made after the Chernobyl accident, as well as extensive refurbishment including replacement of fuel channels, 45-year lifetimes have been achieved. In 2023 RBMK units provided just less than one quarter of Russia's nuclear-generated electricity.

For older RBMK units, service lifetime performance recovery (LPR) operations involve correcting deformation of the graphite stack. After dismantling the pressure tubes, longitudinal cutting of a limited number of graphite columns returns the graphite stack geometry to a condition that meets the initial design requirements. The procedure gives each of these older reactors at least three years' extra operation, and may then be repeated. Leningrad 1 was the first reactor to undergo this over 2012-13, followed by the Kursk units, and then Smolensk.

Most reactors have been uprated. In May 2015 Rosenergoatom said it had completed uprating all VVER-1000 reactors to 104% of rated power, and was starting to take them to 107% using advanced TVS-2M fuel design, starting with Balakovo 4. Earlier, uprating of 5% for VVER-440 (but 7% for Kola 4) had been achieved, and in 2015, Kola 3 went to 107%. The overall cost was

less than RUR 3 billion ($60.5 million), according to Rosenergoatom. The cost of this was earlier put at US$ 200 per kilowatt, compared with $2400/kW for construction of Rostov 2. Rosatom said that at the end of 2016 all 11 VVER-1000 units then operating were at 104% of their original capacity with Rostechnadzor approval.

Rosenergoatom has been investigating further uprates of VVER-1000 units to 107-110% of original capacity, using Balakovo 4 as a pilot plant to 2014. For the V-320 units, pilot commercial operation at 104% power is carried out over three fuel campaigns, with the reactor and other system parameters being monitored and relevant data collected. After this period, a cumulative 104% power operation report is produced for each plant. Rostechnadzor then assesses safety to license commercial operation at the higher power level.

Rosenergoatom is considering the introduction of a 24-month fuel cycle at new nuclear power units. Previously, VVER-1000 reactors operated for 12 months without refuelling and from 2008 they were all converted to an 18-month fuel cycle. VVER-440s still use a 12-month cycle. To achieve 24 months in new units, the design of VVERs will need to be changed and fuel enrichment would need to be increased from 4-4.5% U-235 to 6-7% in the VVER-TOI design.

Individual operating power plants

Balakovo: Rostechnadzor approved a 4% increase in power from all four Balakovo V-320 reactors and major overhauls were undertaken from 2012. Balakovo 1 was upgraded at a cost of RUR 9 billion over nine years, and in December 2015 Rostechnadzor gave it a 30-year operating lifetime extension, the first Russian unit to achieve this. Rosatom has done the same for the other three units, all of which are uprated to 104% with 18-month refuel cycle. All four Balakovo units have now received 30-year lifetime extensions.

Three test assemblies of REMIX fuel were loaded into Balakovo 3 in June 2016. In September 2021 TVEL announced that the trial of the REMIX fuel over five years had been successful. The first six full assemblies of REMIX fuel were loaded into Balakovo 1 in December 2021 for a five-year trial.

Beloyarsk: Beloyarsk 3 BN-600 fast neutron reactor in Zarechny municipality of Sverdlovsk region was upgraded for a 15-year operating lifetime extension, to 2025, and is now licensed to 2040. In 30 years of operation to late 2011, it produced 114 TWh with

capacity factor of 76%. Due to progressive modification, its fuel burn-up has increased from 7% (design value) to 11.4%. It provides heat for Zarechny town as well as electricity from three 200 MWe turbine generators.

The Beloyarsk 4 BN-800 fast neutron reactor was delayed by lack of funds following construction start in 2006 and after first criticality in June 2014 it came online with grid connection in December 2015. Its three steam generators drive a single turbine generator. It entered commercial operation at the end of October 2016. Total cost of construction was reported as RUR 145.6 billion. Despite being a test bed for new fuels, it produced 13.7 TWh in its first 36 months.

Beloyarsk 5 is planned as a BN-1200 reactor (see below).

(Further details on Beloyarsk 4&5 is in the Transition to Fast Reactors and Reactor Technology subsections below.)

Bilibino: Unit 1 was shut down in 2018; units 2-4 were to be shut down in December 2021 after the floating nuclear power plant at Pevek was online, but in 2019 unit 2 was given a licence extension to 2025. Units 2&3 were also granted life extensions to 2025 in September 2023. Decommissioning of unit 1 commenced in January 2019. Units 2-4 were shut down in December 2025. The units were EGP-6 light water graphite-moderated reactors.

Kalinin: Unit 1 had a major overhaul in 2012 for licence extension and power uprate, and Kalinin 2 followed to 2016. Unit 3 was officially approved for this in June 2019. Kalinin 1 was undergoing tests at 104% in 2013 and in mid-2014 it was granted a ten-year licence extension, to mid-2025. In July 2025 its licence was extended by a further 19 years to June 2044. Kalinin 2 licence extension to 2038 was granted in November 2017. All four Kalinin units are set for a 60-year operating lifetime.

Kalinin 4 is a V-320 unit built by Nizhny-Novgorod Atomenergopoekt. Rostechnadzor approved an operating licence in October 2011, it started up in November, was grid-connected in December and attained full commercial operation in September 2012. It uses major components originally supplied for Belene in Bulgaria. Final cost was RUR 7 billion ($220 million) under budget – about 10%. Silmash (Power Machines) upgraded the turbine generator of units 3&4 to increase their gross power to 1100 MWe in 2016.

Preparing Kalinin 4 for startup (Rosatom)

Kola: Kola 3&4 have received licence extensions to 2027 and 2029 respectively. Both units have been uprated to 107%. Low power demand in the Murmansk region and Karelia means they are not fully utilized.

Kola 1&2 have each received second 15-year licence extensions to 2033 and 2034.

Annealing of the unit 2 reactor vessel was undertaken in 2016, with the service life extended to the end of 2029. Work to extend the operating period of unit 1 to 60 years started in 2016, and in July 2018 it was granted a licence extension to 2033. It was reconnected to the grid in December 2018 after a 250-day shutdown, including annealing of the reactor vessel. Further work was undertaken on unit 2 in 2019 during a 268-day outage, including the introduction of new active and passive safety systems, enhancement of the seismic stability of reactor equipment, and replacement of radiation monitoring, control, reactor protection and diagnostic systems. In December 2019 it was granted a licence extension to 2034. Previous annealing of units 1&2 had been undertaken in 1989, followed by further major works over 1991 to 2005, costing $718 million. Some $96 million of this was from international sources including neighbouring countries.

The Kola reactors will be the first VVERs to run on reprocessed uranium (RepU) as a matter of course.

Kursk: Having had a licence extension to 2016, Kursk 1 was the first RBMK unit to be licensed for pilot operation with 5% uprate

(reported to 1020 MWe net) but units 2&4 were also operating at this level late in 2011. In February 2012 Rosatom said it would invest a further RUR 30 billion in upgrading Kursk 2-4 and extending their operating lives – RUR 5.0, 11.9 & 13.7 billion respectively.

On units 1&2 work on the graphite moderator stack was undertaken to avoid the deformation experienced in Leningrad 1. Unit 2 was returned to service in February 2014 after its 'lifetime performance restoration program'. Following inspection, further work was postponed for unit 1, and was then completed in April 2016. Kursk 4 was issued a 15-year licence extension to December 2030 after RUR 13 billion in upgrade work over ten years. Kursk 1 was shut down in December 2021, and Kursk 2 was shut down in January 2024 – both after 45 years in operation.

Units 3-4 are each set for 45-year operating lifetimes.

Kursk 5 – an upgraded RBMK design – was more than 70% built before the project was terminated. Rosatom was keen to see it completed and in January 2007 the Duma's energy committee recommended that the government fund its completion by 2010. However, funds were not forthcoming and the economic case for completion was doubtful, so in February 2012 Rosatom abandoned the project. Instead, major announcements were made regarding Kursk II (see below).

Leningrad: In 2010, intended life extension was announced for Leningrad 4 (15 years), and it underwent a RUR 17 billion refurbishment over 2008-11, including replacement of generator stator. The upgrading investment in all four Leningrad RBMK units totalled RUR 48 billion to early 2012. Leningrad unit 1 was shut down in May 2012 due to deformation of the graphite moderator, and after a RUR 5 billion restoration of the graphite stack as the pioneer lifetime performance recovery (LPR) procedure it was restarted in November 2013. The same work was undertaken on unit 2 in 2014.

Unit 1 was finally shut down in December 2018, followed by unit 2 in November 2020. Defuelling of unit 2 was completed in December 2024. Units 3&4 have been granted licence extensions to 2030. Two VVER-1200 units of phase II of the project are in operation, and a third is under construction. One further unit is planned at the site.

Novovoronezh: Units 3&4 gained 15-year licence extensions to 2016 and 2017, then unit 4 was given a further 15-year licence extension, using parts from the shutdown unit 3. They were the first

VVER-440 units to have their operational life extended by annealing the reactor pressure vessels. In 2018 the emergency core cooling system of unit 4 was supplemented so that a drop in primary circuit pressure will automatically release water with boric acid into the core.

A plan for refurbishment, upgrade and life extension of Novovoronezh 5 was announced in mid-2009, this being a prototype of the second-generation VVER-1000 design. The initial estimate was RUR 1.66 billion but this eventually became RUR 14 billion. The 12 months of work from September 2010 included the total replacement of the reactor control system and 80% of electrical equipment, and fitting upgraded safety systems, in particular, those of emergency core cooling and feedwater, and emergency power supply. The unit is now licensed to 2035.

Unit 6 (also referred to as Novovoronezh II-1), the first of a new generation of 1200 MWe class reactors, was grid-connected in August 2016 after about eight years' construction, and unit 7 (unit II-2) was grid-connected in April 2019.

Rostov: In September 2009 Rostechnadzor approved an operating licence for Rostov 2; it started up in January 2010, was grid connected in March, and entered commercial operation in October 2010. It was approved for 104% of nominal power in October 2012, to about 1075 MWe gross. Accident tolerant fuel (ATF) has been in testing in Rostov 2 since 2021. In September 2024 the third 18-month cycle of ATF began.

Rostov 3&4 are effectively new V-320 units. Unit 3 construction restarted in September 2009. It started up and was grid connected in December 2014, reached full power in July 2015, and entered commercial operation in September 2015. In December Rostechnadzor approved a power increase to 104% of the rated level. Unit 4 construction started in June 2010. It started up late in 2017, was grid connected in February 2018 and entered commercial operation in September 2018. See also following section.

Smolensk: Early in 2012 Rosatom announced a RUR 45 billion programme to upgrade and extend the operating lifetime of Smolensk 1-3 RBMK units. At the same time, construction of Smolensk II was to get underway, with the first VVER unit then to come online by 2024 (now expected 2033). In 2012 Smolensk 1 was licensed to December 2022, a ten-year extension after refurbishment. It has since been licensed to 2027. Upgrading unit 2 was undertaken from 2013, and included replacement of fuel

channels and upgrading the reactor control and protection system and radiation monitoring system, as well as reinforcing the building structure. It is licensed to operate until May 2030. Unit 3 upgrade was implemented to March 2019, though it was already operating above 1000 MWe gross. It is licensed to 2034.

New nuclear capacity

There are six reactors currently under construction in Russia.

Under construction reactors in Russia

Reactor name	Model	Reactor type	Net capacity (MWe)	Construction start
Cape Nagloynyn-1	RITM 200S	PWR	50	2022-08-31
Cape Nagloynyn-2	RITM 200S	PWR	50	2022-08-31
Kursk 2-2	VVER-TOI	PWR	1101	2019-04-15
Leningrad 2-3	VVER V-491	PWR	1101	2024-03-14
Leningrad 2-4	VVER V-491	PWR	1101	2025-03-20
Seversk BREST-OD-300	BREST-OD-300	FBR	300	2021-06-08

In June 2022 Strana Rosatom said that the government has set a goal to bring the share of nuclear power in electricity supply to 25%. Rosenergoatom said that would be challenging because some 10 units are due to shut down by 2030. In order to offset these retirements, Rosenergoatom has said it would need to increase efficiency at operating units. Among other measures, it aims to transfer all VVER-1200 power units to an 18-month fuel cycle.

A draft plan released in September 2024 by the Unified Energy System (UES) of Russia proposed that by 2042 the share of electricity generated by nuclear power will have increased to 23.5%. The draft plan includes 34 new nuclear power units, which are listed in the tables of planned and proposed reactors below. In April 2025 the government approved the Energy Strategy of the Russian Federation until 2050, which targets a 25% nuclear share by 2045 and instructs the construction of 38 new nuclear power units.

Earlier plans are detailed in the appendix.

Russia's long-term goal is to move to fast neutron reactors and closed fuel cycle, outlined below in the Transition to Fast Reactors section.

Power reactors planned

Reactor	Reactor type	MWe gross	Deployment
Beloyarsk 5	BN-1200	1220	2034
Khabarovsk 1&2	VVER-600/V-498	2 x 600	2036
Kola II 1-4	VVER-600/V-498	4 x 600	2035
Krasnoyarsk 1-4	VVER-1200/V-510	4 x 1255	2037
Kursk II-4	VVER 1200/V-510	2 x 1255	2031
Novocherkassk 1&2	VVER-1200/V-510	2 x 1200	2036
Primorsk 1&2	VVER-600/V-498	2 x 600	2039
Reftinskaya 1	VVER-1200/V-510	1255	2036
Smolensk II 1&2	VVER-1200/V-510	2 x 1250	2033
South Urals 1&2	BN-1200	2 x 1220	2038
Ust-Kuyga, Yakutia 1&2	RITM-200N	2 x 55	2030
Total of 24 planned		22,255 GWe	

VVER-1200/V-510 is the reactor portion of the VVER-TOI plant. South Urals was to be BN-800, and now is to be BN-1200.

South Ural is at Ozersk, Chelyabinsk region, 140 km west of Chelyabinsk in Sverdlovsk region. Primorsk is in the far east.

Power reactors proposed

Reactor	Reactor type	MWe gross
Cape Nagloynyn, Chukotka	Shelf-M	10
MPEB Baimsky Chukchi 1-4	RITM-200S	4 x 106
Norilsk ASMM 1-4	RITM-400	4 x 80

Rostov 2, 3&4 (formerly Volgodonsk)

The environmental statement and construction application were approved by Rostechnadzor in May 2009, the construction licence

was granted to Energoatom in June, and construction resumed about September (it had started in 1983). First new concrete for unit 4 was in June 2010. The plant is 13.5 km from the city on the banks of Volgodonsk Tsimlyansk reservoir. Rosatom brought forward the completion dates of the two units after deciding that they would have V-320 type of VVER with improved steam generators and capacity of 1100 MWe. This was expected to save some RUR 10 billion relative to the AES-2006 technology, as it continued the construction done over 1983-86.

OMZ's Izhorskiye Zavody facility at Izhora provided the pressure vessel for unit 3. Nizhniy Novgorod Atomenergoproekt (now NIAEP-ASE) was principal contractor for units 3&4, expected to cost 130 billion according to Rosenergoatom in August 2012. Steam generators for unit 4 were from AEM-Tekhnologi at the Atommash plant, those for unit 3 from ZiO-Podolsk. Ukraine's Turboatom provided the low-speed turbine generators for both units. Grid connection of unit 2 was in March 2010 and full commercial operation was in October. Unit 3 started up and was grid-connected in December 2014, and entered commercial operation in September 2015. Unit 4 started up in December 2017, was grid-connected five weeks later in February 2018 and entered commercial operation in September 2018. The Rostov power plant supplies Crimea, annexed by Russia in 2014.

Novovoronezh II

The principal contractor for Novovoronezh Phase II is JSC AtomEnergoProekt (Moscow), with work starting in 2007 and some involvement of NIAEP-ASE. Construction is now under the ASE group. This was the lead plant for deploying the V-392M version of the AES-2006 units. First concrete was poured for unit 1 (the 6th unit at the site) in June 2008 and for unit 2 in July 2009. Unit 1 was initially expected to be commissioned in 2015, with unit 2 following a year later, at a total cost of US$ 5 billion for 2228 MWe net (1114 MWe net each). The reactor pressure vessels are from OMZ Izhora and the advanced steam generators from ZiO-Podolsk, with 60-year operating lifetime expectancy. Turbine generators (high speed) are from Power Machines.

Atomenergoproekt told its contractors in December 2014 to accelerate work, but in May 2015 a delay of one year in commissioning both units was announced, due to low power demand. In September 2015 a pre-startup peer review was conducted for unit 1 under World Association of Nuclear Operators

(WANO) auspices. Rostechnadzor issued the operating licence for unit 1 in March 2016, and fuel loading commenced. It started up in May and was grid-connected in August 2016. Unit 2 was due to enter commercial operation in January 2019, but in February 2018 Rosenergoatom announced that it would slow construction in response to slowing demand and pressure from power consumers to reduce rate increases. Fuel loading was completed in February 2019, with grid connection in May, and commercial operation in October 2019. The plant is on one of the main hubs of the Russian grid.

In July 2020, Rosenergoatom announced that unit 1 would switch to an 18-month refuelling cycle (from 12 months) for a trial period of about three years.

Leningrad II

A general contract for Leningrad phase II AES-2006 plant was signed with St Petersburg Atomenergoproekt (SPb AEP, merged with VNIPIET to become Atomproekt) in August 2007 and Rostechnadzor granted site licences in September 2007 for two units. A specific engineering, procurement and construction contract for the first two V-491 units was signed in March and Rostechnadzor issued a construction licence in June 2008. First concrete was poured on schedule for unit 1 in October 2008 and it was due to be commissioned in October 2013. However, a section of outer containment collapsed in 2011 and set back the schedule, as did subsequent manpower shortage, so that commissioning was then expected in 2016, following start-up at the end of 2015. Rostechnadzor granted a construction licence for the second reactor in July 2009, and first concrete was poured in April 2010. Commercial operation was due in 2018 but in May 2015 a delay of one year in commissioning both units was announced, due to low power demand. Unit 1 achieved first criticality in February 2018, and was grid connected in March, with Rostechnadzor approving commercial operation by October 2018. Rosenergoatom then announced a delay to the start of commercial operation of unit 2 to 2020. The delay was requested by energy consumers to reduce rate increases. In July 2020 Rosatom reported that first fuel assemblies had been loaded, and the reactor was connected to the grid in October 2020. Unit 2 entered commercial operation in March 2021 following a delay due to measures to limit the spread of coronavirus. Each reactor will also provide 1.05 TJ/h (9.17 PJ/yr) of district heating. They are designed to replace the oldest two Leningrad units, which shut down in 2018 and 2020.

The 2008 construction contract was for $5.8 billion ($2480/kW) possibly including some infrastructure. Total project cost was estimated at $6.6 billion. In May 2015 Titan-2 became general contractor for units 1&2*, with Atomproekt remaining the general designer, and in October 2015 Titan-2 became also the principal equipment supplier. Construction is now under the ASE group which consolidates most of the entities involved.

* It was reported in September 2011 that Titan-2, a major subcontractor, took over from SPb AEP as principal construction contractor, then in February 2012 that Spetsstroy of Russia (Federal Agency for Special Construction) would do so. In December 2013 Roesenergoatom transferred the project from Spetsstroy to Atomenergoproekt Moscow as principal contractor, while SPb AEP/VNIPIET/Atomproekt remained architect general. NIAEP-ASE also bid for the general contract in October 2013. Rosatom had said in February 2012 that it did not believe that SPb AEP should perform the full range of design, construction and equipment supply roles.

A design contract for the next two units (3&4) was signed with SPb AEP in September 2008, and public consultation on these was held in Sosnovy Bor in mid-2009. An environmental review by Rostechnadzor was announced for them in January 2010 and site development licences were granted in June, then renewed in April 2013. Rosenergoatom signed a contract with VNIPIET at the end of December 2013 to develop project documentation. It expected construction licences in 2014 and construction start in 2015, but the delay to units 1&2 extended to units 3&4.

Construction of Leningrad II-3 commenced in March 2024, followed by Leningrad II-4 in March 2025.

Nizhny Novgorod

The plant in Navashino District near Monakovo was to comprise four AES-1200 units of 1150 MWe net and cost RUR 269 billion. The first was originally planned to come online by 2019 to address a regional energy deficit. In February 2008 Rosatom appointed Nizhny-Novgorod Atomenergoproekt (NN-AEP or NIAEP) as the principal designer of the plant. Rostechnadzor issued a positive site review for units 1&2 early in 2010 and a site licence with prescription for site monitoring in January 2011. Rosatom's proposal to proceed with construction of two units was approved in November 2011. Site works started in 2012 and formal construction starts were expected soon after. This was to be the first VVER-TOI plant, with rated capacity of 1255 MWe per unit. In

the government decree of August 2016 two VVER-TOI were specified, for completion by 2030. However the plant was not included in Russia's draft energy plan, released in September 2024.

Tatar

A 4000 MWe nuclear plant was under construction and due on line from 1992, but construction ceased in 1990. Then a two-unit VVER-1200 plant was included in the Regional Energy Planning Scheme in November 2013. In the government decree of August 2016 a single VVER-TOI was listed for completion by 2030 at Kamskiye Polyany in Nizhnekamsk Region of Tatarstan. Russia's draft energy plan in September 2024 did not list a plant at Tatar.

Central/Kostroma

The 2340 MWe Tsentral (Central) nuclear power plant was to be 5-10 km northwest of Buisk Town in the Kostroma region, on the Kostroma River. It was another of those deferred but following Rosatom's October 2008 decision to proceed, it appeared that construction might start in 2013. Rostechnadzor approved the site and a development licence was expected by mid-2010, then a construction licence in 2012. A two-unit VVER-1200 plant was included in the Regional Energy Planning Scheme in November 2013, with both VVER-TOI units to be online by 2030. Moscow Atomenergoproekt was to be the architect-engineer. However, the plant was not in Russia's draft energy plan released in September 2024.

South Urals

The Yuzhnouralskaya plant near Ozersk in Chelyabinsk region has been twice deferred, and was then reported by local government to have three BN-1200 fast reactor units planned, instead of four VVER-1200. Then a two-unit BN-1200 plant was included in the Regional Energy Planning Scheme in November 2013. Plans for an initial BN-1200 unit were confirmed in August 2016, for completion by 2030. There is only enough cooling water (70 GL/yr) for two of them, and the third will depend on completion of the Suriyamskoye Reservoir. The plant was not included in Russia's draft energy plan, released in September 2024.

Kola II

In January 2012 Rosenergoatom said that the replacement Kola II plant, about 10 km south of the present plant in the Murmansk region and on the shores of Lake Imandra, would be brought forward and built with two VVER-TOI units to come on line in 2020. Then a two-unit VVER-1200 plant was included in the Regional Energy Planning Scheme in November 2013; but in September 2014 Rosenergoatom was considering medium-sized units, either VVER-600 or VBER-600 for Kola. In the government decree of August 2016 a single VVER-600 was specified, for completion by 2030. In June 2021 the plant management announced that construction would start on two VVER-S-600 reactors for Kola II in 2028, with the first to be online in 2034. The 'S' signifies spectral shift control, with heavy water in the primary coolant. Russia's draft energy plan, released in September 2024, showed three VVER-600 units starting construction between 2035 and 2040. In 2025 Rosatom confirmed plans for four VVER-S-600 units at Kola II, with construction start in 2028 and the first unit online in 2034.

Kursk II

It was originally envisaged that the first unit of Kursk II should be online by the time Kursk 1 closes, then envisaged in 2016. In March 2011, the State Duma's Energy Committee recommended that the government update the general scheme of deployment of electricity generators, to have units 1 and 2 of Kursk II being commissioned in 2020 and 2023 as the lead project with VVER-TOI types, and Kursk II-1 being the reference unit for VVER-TOI. Kursk I-5 capacity had been planned in the federal target programme and its abandonment left a likely base-load shortfall for UES in central Russia.

Rosatom started engineering surveys for Kursk II in 2011, and set up a task force of representatives from the nuclear industry and Kursk Region government to produce project documentation on construction of Kursk II. In June 2012 Rosatom appointed Moscow AEP as designer, and Nizhny-Novgorod AEP (NIAEP) as architect general and principal contractor.Site work commenced at the end of 2013, with environmental assessment. In June 2016 Rostechnadzor issued a construction licence to Rosenergoatom for unit 1, and the main site works commenced later that month. A licence for unit 2 was issued in October 2016. Atommash supplied the reactor pressure vessels and steam generators, Power Machines the

turbine generators, and Energoteks the core catcher. Construction of Kursk II-1 started in April 2018, and unit 2 in April 2019. Kursk II-1 achieved first criticality in May 2025 and was connected to the grid in December 2025. It reached full power in March 2026, with commercial operation expected later in 2026.

Earlier, in March 2025, Rostechnadzor issued site licences for Kursk II-3&4. First concrete was poured for Kursk II-3 in January 2026.

Smolensk II

Atomenergoproekt Moscow was architect engineer for VVER-TOI units to replace old RBMK capacity at Smolensk. Rosenergoatom's investment concept was approved in 2011. Site surveys were undertaken from June 2013, and three potential sites were shortlisted. In mid-2017 a special decree was issued for the purchase of 400 ha and for site works 6 km from Smolensk I. A four-unit VVER plant was included in the Regional Energy Planning Scheme in November 2013, with two units on line by 2025 and two by 2030. Engineering surveys were completed in November 2014 at Pyatidvorka (6 km from Smolensk I). Construction start was then deferred to 2022, with the first unit expected online in 2027. Rostechnadzor was expected to issue a site licence in September 2016. Russia's draft energy plan, released in September 2024, shows Smolensk II-1&2 as VVER-TOI units due to start construction in 2033 and 2035 respectively. Site preparation and infrastructure construction began in 2025.

Seversk

The first 1200 MWe unit of the Seversk AES-2006 plant 32 km northwest of Tomsk was due to start up in 2015 with the second in 2017. A two-unit VVER-1200 plant was included in the Regional Energy Planning Scheme in November 2013, but did not proceed. both units to be on line by 2030. The plan was for plant to supply 7.5 PJ/yr of district heating. Seversk is the site of a major enrichment plant and former weapons facilities.

Atomenergopoekt Moscow was to build the plant at estimated cost of RUR 134 billion ($ 4.4 billion). Rostechnadzor granted a site development licence in November 2009 and a further site licence in 2011. Provisional site work had commenced, and a design contract for the low-speed turbine generators had been signed between

Moscow AEP which was responsible for design and engineering, and Alstom Atomenergomash.

This would be the first Russian plant using the low-speed turbines.

In 2019 Seversk was put on the updated general scheme of deployment of energy facilities, with the first reactor commissioning before 2020 and the second one 2020-2025. However, in the government decree of August 2016 a single BREST-300 fast reactor was the only unit specified, which began construction in June 2021. Russia's draft energy plan, released in September 2024, did not show any further units planned at Seversk.

Baltic

Separately from the February 2008 plan, Rosatom energy-trading subsidiary InterRAO UES proposed a Baltic or Baltiyskaya AES-2006 nuclear plant in Kaliningrad on the Baltic coast to generate electricity for export, and with up to 49% European equity. Private or foreign equity would be an innovation for Russia. The plant was designed to comprise two 1200 MWe VVER units, V-491 model, sited at Neman, on the Lithuanian border and costing some RUR 194 billion (in 2009 value, €4.6 billion, $6.8 billion), for 2300 MWe net. Project approval was confirmed by government decree in September 2009, following initial approval in mid-2008 as an amendment to the federal target program (FTP) of 2007. The mid-2011 business plan estimated the likely capital cost to be €6.63 to 8.15 billion.

WorleyParsons was appointed technical consultant for the project. Rosenergoatom set up a subsidiary: JSC Baltic NPP to build and commission the plant. St Petersburg Atomenergoproekt - VNIPIET (now merged as Atomproekt) was to be the architect engineer, Nizhniy Novgorod AEP (NIAEP) was to be construction manager, with Atomstroyexport (ASE). TitanStroyMontazh was to be engineering subcontractor. Originally AEM Petrozavodskmash was to produce the pressure vessel for unit 1 but this was assigned to AEM-Tekhnologii at the Atommash plant. OMZ's Ishorskiye Zavody was to produce the pressure vessel for unit 2 and the pressurizers for both units. Alstom-Atomenergomash was to supply the Arabelle low-speed turbine generators for both units – the plant was to be the JV's first customer, and the Baltic plant was to be the first Russian plant to use major foreign components. (LMZ high-speed turbine generators had initially been approved.)

Site work began in February 2010. Expenditure to January 2012 was RUR 7.25 billion ($241 million), and that in 2012 was expected to be RUR 7 billion. Rostechnadzor issued a construction licence for unit 1 in November 2011 and first concrete was poured on (revised) schedule in April 2012, with the base completed in December 2012. Unit 1 was planned to come on line in October 2016, after 55 months construction, supplying Rosenergoatom. Commercial operation was due in 2017. Second unit construction was planned over 2013-18, with 48 months to first power and full operation in April 2018. NIAEP-ASE suspended construction in June 2013 (see below), pending a full review of the project intended to be by mid-2014, though some work on the containment was ongoing in following months. Rosenergoatom said that in October 2013 it had spent RUR 50-60 billion ($1.2 to 1.6 billion) on the project. In 2017 the pressure vessel made for unit 1 was sent to Ostrovets in Belarus, replacing one that had suffered a mishap there.

InterRAO UES was responsible for soliciting investment (by about 2014, well after construction start) and also for electricity sales. The Baltic plant would have directly competed with the unit that was planned at Visaginas near Ignalina in Lithuania and with plans for new nuclear plants in Belarus and Poland. Rosenergoatom said that the plant was deliberately placed "essentially within the EU" and was designed to be integrated with the EU grid. Most of the power (87% in the mid 2011 business plan) was to be exported to Germany, Poland and Baltic states. Transmission to northern Germany was to be via a new undersea cable, and in 2011 Inter RAO and Alpiq agreed to investigate an 800 MWe undersea DC link to Germany's grid. Lithuania was invited to consider the prospect, instead of building Visaginas as a Baltic states plus Poland project, but declined.

* Lithuania's revised energy policy in 2012 involved rebuilding its grid to be independent of the Russian/Belarus system and to work in with the European Network of Transmission System Operators (ENTSO) synchronous system, as well as strengthening interconnection among the three Baltic states.

Czech power utility CEZ earlier expressed interest in the project, as did Iberdrola from Spain, whose engineering subsidiary already has worked at Kola, Balakovo and Novovoronezh nuclear power plants. In April 2010 Enel signed a wide-ranging agreement with Inter RAO which positioned it to take up to 49% of the plant, but this did not proceed. Rosatom earlier said that the project would not be delayed even if 49% private equity or long-term sales contracts were not forthcoming.

However, in June 2013 construction was suspended due to lack of interest in the project from the Baltic states, Poland and Germany. Construction has remained stalled since then, but some manufacture and supply of equipment continued. The polar crane was delivered in August 2014, and a contract for storing four steam generators for 15 months from July 2015 was let for RUR 46 million.

In July 2015 Kaliningrad local government was talking up the prospects of an aluminium smelter to justify resuming construction of the plant. However, in September 2015 the first deputy director general for operations management at Rosatom said that only when long-term electricity sales contracts are negotiated and "formalized in binding documents, *i.e.* contracts for buying electricity produced by plant from the western side, we could speak of continuation of construction."

In April 2017 it was confirmed that the RPV fabricated for Baltic 1 would be sent to Belarus for Ostovets 2. See also grid implications in Electricity Transmission Grids information page.

Other reactor designs have been considered at Kalingrad. NIAEP said in 2013 it was investigating building some small nuclear plants in Kaliningrad instead – eight 40 MWe units such as those on floating nuclear power plants was mentioned as a possibility, and they would fit into the local energy system better, with its 500 MWe total requirement. In mid-2014 Rosenergoatom was considering a VVER-600 from Gidropress with many of the same components as the original VVER-1200, and a VBER-600 from OKBM, the latter being less developed so involving a two-year delay. A new schedule and site configuration, involving small units, was to be approved by mid-2014, but this did not proceed.

The 2015 Rosenergoatom annual report said: "Rosatom State Corporation has recently updated the concept for the Baltic NPP project implementation and is supposing to supply up to 100% of power outside the Kaliningrad Oblast. As part of ensuring the technical capability to supply power, several options for the power generation pattern of the Baltic NPP are being studied, taking into account future configuration of synchronization zones, and the Kaliningrad Oblast plans to prepare for an isolated mode of operation. As part of ensuring commercial conditions of supplies, the negotiations with potential buyers of electricity in the EU countries are continued. So far, several memoranda of understanding, and electricity purchase and sale agreements have been signed with major European energy holdings."

As well as the Baltic plant, two other ventures with Rusal (see below) were to require private equity, but these did not proceed.

Tver

The plant at Udomlya district and 4 km from Kalinin was being designed by Nizhny-Novgorod Atomenergoproekt (NN-AEP), and in January 2010 it was announced that Rostechnadzor would conduct an environmental review of it for the first two VVER-1200 units, these being on the general scheme of electricity generators deployment to 2020. No firm dates have been given for the project, though a site development licence was expected in March 2010.

Pevek

Energoatom signed a RUR 9.98 billion purchase contract for the first floating nuclear power plant then intended for Vilyuchinsk, on the Kamchatka Peninsula in the Far East, in 2009. Keel-laying took place in April 2007 at the Seymash shipyard in Severodvinsk on the White Sea, but the project was then transferred to the Baltiyskiy Zavod shipyard at St Petersburg. A second keel-laying took place in May 2009 at Baltiyskiy Zavod shipyard. The 2x35 MWe plant, named *Academician Lomonosov*, was due to be commissioned in 2012, but the project was delayed due to shipyard insolvency. In 2012 the plant was re-assigned to Pevek in the far northeast of Siberia. The twin reactors commenced operation in December 2019, with commercial operation in May 2020. See FNPP subsection below.

Transition to fast reactors

It is envisaged that fast neutron power reactors will play an increasing role in Russia, with substantial recycle of fuel.

Russia envisages full recycling of fuel, balancing thermal and fast reactors, so that 100 GWe of total capacity requires only about 100 tonnes of input per year, from enrichment tails, natural uranium and thorium, with minor actinides being burned. About 100 t/yr of fission product waste would go to a geological repository.

The sodium-cooled BN-series fast reactor plans are part of Rosatom's *Proryv*, or 'Breakthrough', project to develop fast reactors with a closed fuel cycle whose mixed oxide (MOX) fuel will

be reprocessed and recycled. The BN-600 reactor at Beloyarsk has operated successfully since 1980 and is now licensed to 2040. The BN-800 reactor at Beloyarsk has operated since 2014, essentially as a demonstration unit for fuel and design features for the BN-1200, which is now deferred.

Recent priority in financing has been for lead-cooled fast neutron reactors with dense nitride fuel. Initially two projects were proposed – the BREST-300 lead-cooled fast reactor with associated nitride fuel fabricating/re-fabricating and spent fuel reprocessing facilities and the SVBR-100 lead-bismuth fast reactor, since dropped. Hence from the mid-2020s, fast reactors will be new designs such as BREST with a single core and no blanket assembly for plutonium production. The first BREST-OD-300 reactor commenced construction in June 2021 at Seversk.

Fast reactors represent a technological advantage for Russia. In late 2012 Rosatom said that it would make available its experimental facilities for use as part of the Generation IV International Forum, including large physical test benches at Obninsk's Institute of Physics and Power Engineering, the BOR-60 research reactor at NIIAR, and the multifunction research reactor MBIR that is under construction at the NIIAR site.

While Rosatom plans to invest its own funds into FNR development through to 2025, in October 2018 it asked the government to allocate an additional RUR 200 billion ($3.05 billion) over 2019-2025 under the federal target programme for nuclear power. BREST is the focus of this, and Rosatom's long-term strategy up to 2050 involves moving to inherently safe nuclear plants using fast reactors with a closed fuel cycle and MOX or nitride fuel.

Further details of fast neutron power reactors are in the Reactor Technology section below.

Federal Target Programme: Advanced Nuclear Power Technologies 2010-2020

Rosatom put forward two fast reactor implementation options for government decision in relation to the federal target programme (FTP) 'Advanced Nuclear Power Technologies 2010-2020'. The first focused on a lead-cooled fast reactor such as **BREST** with its fuel cycle, and assumed mobilization of all available resources on this project with a total funding of about RUR 140 billion (about $3.1 billion). The second multi-track option was favoured, since it involved lower risks than the first. It would result in technical

designs of the Generation IV reactor and associated closed fuel cycles technologies by 2014, and a technological basis of the future innovative nuclear energy system featuring the Generation IV reactors working in closed fuel cycles by 2020. A detailed design would be developed for a multi-purpose fast neutron research reactor (MBIR) by 2014 also. This second option was designed to attract more funds apart from the federal budget allocation, was favoured by Rosatom, and was accepted.

In January 2010 the government approved the federal target programme (FTP2010) "New-generation nuclear energy technologies for the period 2010-2015 and up to 2020" designed to bring a new technology platform for the nuclear power industry based on fast neutron reactors. It anticipated RUR 110 billion to 2020 out of the federal budget, including RUR 60 billion for fast reactors, and subsequent announcements started to allocate funds among three types: BREST, SVBR (now dropped) and continuing R&D on sodium cooled types.

The FTP involved plans to build and commission a commercial complex to fabricate dense fuel, to complete construction of a pilot demonstration pyrochemical complex to fabricate BN fuel, and to test closed fuel cycle technologies. Fusion studies are included and the total R&D budget was RUR 55.7 billion, mostly from the federal budget. The FTP2010 implementation was intended to result in a 70% growth in exports of high technology equipment, works and services rendered by the Russian nuclear industry by 2020. It was also to commercialize new fast neutron reactors for Russia to build over 2020-2030. In 2012 the head of Rosatom said that the FTP was being accelerated to bring forward development and have a full range of fast reactor technologies with associated fuel cycles operating by 2020. Rosatom's R&D budget would be almost doubled by then to achieve this.

In March 2017 Rosatom and Russian Venture Company (RVC) signed an agreement to cooperate in the promotion of advanced technologies and innovative developments at Rosatom's subsidiaries. RVC is a Russian state institution responsible for funding national innovation projects on behalf of the National Technology Initiative (NTI), established in 2014. The agreement has wide scope.

Federal target programme 2010 funding for fast neutron reactors to 2020

Cooling	Demonstration reactor	Construction (RUR billion)	R&D (RUR billion)	Total (RUR billion)

Pb-Bi cooled	SVBR 100 MWe	10.153	3.075	13.228
Na cooled	(BN-600, BN-800)	0	5.366	5.366
Pb cooled	BREST 300 MWe	15.555	10.143	25.698
multiple	MBIR 150 MWt	11.390	5.042	16.432
	Total:	37.1		60.7

Source: Government decree #50, 2010. Most (RUR 9.5 billion) of the funding for SVBR was to be from "other sources" than the state budget, and it has now been dropped.

In March 2018 the FTP2010 was amended by the government in the light of reduced energy demand projections.

In December 2018 it was cancelled.

MBIR

Design of the 150 MWt multi-purpose fast neutron research reactor (*mnogotselevoy issledovatilskiy reaktor na bystrych neytronach*, MBIR) was finalized in 2014 by NIKIET and the equipment contract let to Atomenergomash-Technologies. Rostechnadzor issued a site licence in 2014, a construction licence in May 2015, and construction started in September 2015 at the Research Institute of Atomic Reactors (RIAR or NIIAR) in Dimitrovgrad, as part of the Nuclear Innovation Cluster there. The total project cost was then quoted as RUR 40-41 billion, with some of this expected from investors, and completion was scheduled for 2020. However, the project was paused shortly after construction began. In November 2020 Rosatom appointed a new contractor, AO Institut Orgenergostroy, and construction resumed, with commissioning now expected in 2027.

The MBIR will be a multi-loop research reactor capable of testing lead, lead-bismuth and gas coolants as well as sodium, and running on MOX fuel. Initially it will be sodium-cooled. It will be part of an international research centre at RIAR's site and the IAEA was expected to sign an agreement on the MBIR International Research Centre in September 2016. The project is open to foreign collaboration, in connection with the IAEA INPRO programme. In April 2017 Rosatom was soliciting Japanese involvement. The MBIR will replace the BOR-60 fast research reactor. See also R&D section in the information page on Russia's Nuclear Fuel Cycle.

In January 2023 the reactor vessel for the MBIR was set in place marking the last major installation that requires the building to have an open top, with the next step being the construction of the reactor building dome. In August 2024 the first batch of pilot fuel elements for MBIR were produced. Installation of primary circuit equipment began in 2025.

Proryv (Breakthrough) project

The Proryv project is to create a new generation of nuclear power technologies on the basis of a closed nuclear fuel cycle using fast neutron reactors. This is proceeding as a high priority in nine coordinated centres.

The basic concepts include elimination of severe reactor accidents, closing the fuel cycle, low-activity radioactive waste, non-proliferation, reduced capital cost of fast reactors.

The nine responsibility centres include:

- Reprocessing technology and radioactive waste management for the reprocessing module (RM) of the pilot demonstration energy/power complex (PDEC or PDPC).
- Pilot production lines for onsite nuclear fuel cycle, including the fabrication/refabrication module (FRM) and the fast reactor used fuel reprocessing module (RM).
- Development of fuel elements and assemblies with mixed nitride uranium plutonium (MNUP) fuel, at Bochvar National Research Institiute (VNIINM).
- Building and operating the BREST-OD-300 reactor, at JSC NIKIET.
- Development of materials for the BN-1200 fast reactor, at JSC Afrikantov OKBM.
- Design engineering the pilot demonstration energy/power complex (PDEC or PDPC) including nitride fuel fabrication and recycling, and developing an industrial energy complex (IEC).

Most of these initiatives are more fully described in the companion page on Russia's Nuclear Fuel Cycle. BREST and BN-1200 are described below under Reactor Technology.

Floating nuclear power plants

Rosatom was planning to build seven or eight floating nuclear power plants (FNPPs) by 2015. The first of them was to be constructed and then remain at Severodvinsk with intended completion in 2010, but plans changed. Each FNPP was to have two 35 MWe KLT-40S nuclear reactors. (If primarily for desalination this set-up is known as APVS-80.)

A decision to commit to building a series was envisaged to be in 2014 when the first was expected to be near commissioning. Rosenergoatom earlier signed an agreement with JSC Kirov Factory to build further units, and Kirov subsidiary Kirov Energomash was expected to be the main non-nuclear contractor on these.

The keel of the first floating nuclear power plant (FNPP), named *Akademik Lomonosov*, was laid in April 2007 at Sevmash in Severodvinsk, but in August 2008 Rosatom cancelled the contract (apparently due to the military workload at Sevmash) and transferred it to the Baltiysky Zavod shipyard at St Petersburg, which has experience in building nuclear icebreakers. After signing a new RUR 9.98 billion contract in February, new keel-laying took place in May 2009. The 21,500 tonne hull (144 metres long, 30 m wide) was launched at the end of June 2010.

Completion and towing to the site had been expected in 2012 and grid connection in 2013, but due to insolvency of the shipyard JSC Baltijsky Zavod* and ensuing legal processes it was delayed considerably. Barely any work was done over 2011-12 after some RUR 2 billion allocated to finance the construction apparently disappeared. The state-owned United Shipbuilding Corporation acquired the shipyard in 2012 and a new contract with Baltijsky Zavod-Sudostroyeniye (BZS), the successor of the bankrupt namesake, was signed in December 2012. The cost of completing the FNPP was then put at RUR 7.631 billion.

* a subsidiary of privately-owned United Industrial Corporation.

Earlier at the end of 2012 the Ministries of Defence, Energy and Industry agreed to make Pevek the site for the first FNPP unit, the

Academician Lomonosov. Vilyuchinsk, Kamchatka peninsula, was the site originally planned for its deployment to ensure sustainable electricity and heat supplies to the naval base there. Pevek on the Chukotka peninsula in the Chaun district of the far northeast, near Bilibino, was originally planned as the site for the second FNPP, to replace the Bilibino nuclear plant and a 35 MWe thermal plant as a major component of the Chaun-Bilibino industrial hub. Rosenergoatom said that the tariff revenue of Chukotka made it more attractive than the Vilyuchinsk naval base.

The two KLT-40S reactors from OKBM Afrikantov were installed in October 2013. The KLT-40S is a version of the icebreaker reactor for floating nuclear power plants which runs on low-enriched uranium (<20%) and hence has a bigger core and shorter refuelling interval: 3-4.5 years. Operational lifetime is 40 years.

In September 2015 Rosatom signed a cooperation agreement with the government of the Chukotka Autonomous District for power sector development around the Chaun-Bilibino Energy Hub, including installation of the first FNPP at Pevek. Construction of onshore facilities for the plant commenced in September 2016.

Total estimated costs for Pevek increased to RUR 37 billion at May 2015, partly due to factoring in the required site works and infrastructure. The government contributed to coastal infrastructure, with RUR 5 billion over 2016-20. Rosenergoatom said the pilot FNPP was costing Rosenergoatom RUR 21.5 billion, and it expects the second one to be about RUR 18 billion.

Mooring tests started in mid-2016, and in May 2018, the vessel completed the first leg of its journey to Pevek, mooring in Murmansk for fuel loading*. Fuel loading was completed in October 2018, with startup in December 2019, and commercial operation in May 2020. The plant is connected to the region's grid, heat and water supply.

* The Law of the Sea does not address non-propelled nuclear power plants at sea, so concerns by neighbouring countries en route cannot be resolved if reactors are commissioned before they are back in Russian waters

Plans for Floating Nuclear Power Plants

** Initial choice, later changed to Saint Petersburg*

The third site for FNPP deployment was to be Chersky or Sakha in Yakutia. As of early 2009, four floating plants were designated for northern Yakutia in connection with the Elkon uranium mining project in southern Yakutia, and in 2007 an agreement was signed with the Sakha Republic (northeast Yakutia region) to build one of them, using smaller ABV-6* reactors. Five were intended for use by Gazprom for offshore oil and gas field development and for operations on the Kola peninsula near Finland and the Yamal peninsula in central Siberia. The draft energy plan released in September 2024 by UES showed that two land-based RITM units (RITM-200N) are now planned at Yakutia instead.

* The small Russian ABV-6M integral PWR is 16-45 MW thermal. A design known as the Volnolom FNPP consists of a pair of reactors (12 MWe in total) mounted on a 97-metre, 8700 tonne barge plus a second barge for reverse osmosis desalination (over 40,000 m^3/day of potable water).

Earlier in July 2017 Rosatom announced the second generation of FNPPs, now called **optimized floating power units** (OFPUs), would use two RITM-200M reactors derived from those for the latest icebreakers. These are more powerful than the KLT-40S reactors, at 50 MWe each, have fuel enriched to almost 20%, need refuelling only every 10-12 years at a service base, so no onboard used fuel storage is required. Also, the actual reactors are lighter, so the barge is smaller* and displacement is reduced from about 21,000 to 12,000 tonnes. Operational lifetime is 40 years, with possible extension to 60 years.

* Apparently 115m long and 25m wide.

Rosatom is planning four OFPUs hosting eight RITM-200 units at Cape Nagloynyn to supply 440 MWe to the Baimskaya copper mining project south of Bilibino and Pevek. One OFPU will be held in reserve to cover refuelling and maintenance downtime. In September 2021 Rosatom subsidiary FSUE Atomflot and KAZ Minerals subsidiary GDK Baimskaya LLC signed an agreement for electricity supply from four 106 MWe OFPUs. The Rosatom commitment is over RUR 150 billion ($2.1 billion) and the electricity price is RUR 6.45/kWh (¢8/kWh) indexed for inflation. In September 2021 Rosatom awarded a $226 million contract to Wison (Nantong) Heavy Industries in China for the first two 19,100 tonne barge hulls. The reactors and turbines from Atomenergomash will be installed in Russia, at Baltijsky Zavod Shipbuilding in St Petersburg, where the other two barge hulls will be built (that shipyard is committed to building LK-60 icebreakers). In early September 2022 Atomenergomash signified the start of full construction of the first of four OFPUs through a keel laying ceremony.

As well as FNPPs, NIKIET is developing a sunken power plant which will sit on the sea bed supplying electricity for Arctic oil and gas development. This is SHELF, a 6 MWe integral PWR. NIKIET has also proposed its use for the RUR 100 billion Pavlovsky lead-zinc mine project in northern Novaya Zemlya. See also R&D section in the information page on Russia's Nuclear Fuel Cycle.

Export potential

In May 2014 the China Atomic Energy Authority (CAEA) signed an agreement with Rosatom to cooperate in construction of floating nuclear cogeneration plants for China offshore islands. These would be built in China but be based on Russian technology, and possibly using Russian KLT-40S reactors. This arrangement appears to have been displaced by indigenous Chinese developments. In August 2015 Rosatom and Indonesia's BATAN signed a cooperation agreement on construction of FNPPs, but nothing further has been announced.

In June 2024 the Republic of Guinea and Rosatom signed an MOU to cooperate on the development of floating power units to supply electricity in the African country.

Heating

From the 1970s there have been plans for nuclear district heating plants (AST). The AST-500 integral PWR-type plant was designed by OKBM and built by Atommash, with the first one installed at Gorky and ready to start up in September 1989. However, local opposition prevented its operation. (Gorky is now Nizhny Novgorod.)

In the 1990s, 5 GW of thermal power plants (mostly AST-500 integral PWR type) were planned for district and industrial heat to be constructed at Arkhangelsk (four VK-300 units commissioned to 2016), Voronezh (two AST-500 units 2012-18), Saratov, Dimitrovgrad and (small-scale, KLT-40 type PWR) at Chukoyka and Severodvinsk. Also the main Russian nuclear plants were expected to provide 30.8 PJ cogeneration district heating by about 2010. (A 1000 MWe reactor produces about 95 PJ per year internally to generate the electricity.)

A RUTA-70 low-temperature 20-70 MWt pool-type district heating reactor has been designed by NIKIET and proposed for Obninsk. It would operate at atmospheric pressure and would have a subsidiary function as a neutron source for the Institute for Physics and Power Engineering (IPPE) or for desalination by MED – but has evidently not proceeded.

In 2016 four Russian cities expressed an interest in using small reactors to supply heat and power, according to NIKIET. A specific feasibility study was undertaken for an Arkhangelsk nuclear cogeneration plant. A broader Rosatom feasibility study on regional power concluded that up to 38 cogeneration reactors could be deployed at 14 sites for this purpose: Arkhangelsk (4, with high local support), Ishevsk (2), Ivanovo (2), Kazan (3), Khabarovsk (4), Komsomolsk-on-Amur (3), Kurgan (2), Murmansk (2), Perm (2), Tver (2), Ufa (2), Ulyanovsk (3), Vyatka (2) and Yaroslavl (3).

The basic cogeneration plant proposed by NIKIET is twin VK-300 units each rated 250 MWe, or 150 MWe plus 1675 GJ/h, conservatively to give joint annual output of 3 TWh and 16 PJ very competitively.

Many operating nuclear power plants in Russia provide district heat supply.

Heavy engineering and turbine generators

Atomenergomash (AEM) under Rosatom now claims to be the leading company in Russia for major components of nuclear power

plants, controlling over 40 facilities and to be the sole Russian source of steam generators and primary coolant pumps for nuclear plants. AEM-Technology has two main plants: Petrozavodskmash in Karelia and Atommash at Volgadonsk. Atommash has a 15,100-tonne press and is producing reactor pressure vessel forgings for VVER-TOI reactors.

OMZ subsidiary Izhorskiye Zavody produces forgings for nuclear reactors including reactor pressure vessels, steam generators, and heavy piping. In 2008 the company rebuilt its 12,000 tonne hydraulic press, claimed to be the largest in Europe, and a second stage of work increased that capacity to 15,000 tonnes.

In May 2012 Rosenergoatom said that reactor pressure vessels for its VVER-TOI reactors would be made by both Izhorskiye Zavody and the Ukrainian works Energomashspetsstal (EMSS) with Russian Petrozavodskmash. Since then Atommash has also been producing pressure vessel forgings for VVER-TOI reactors. In mid-2015 Rosatom announced that a new nickel-alloy steel coupled with larger (4.2m) diameter pressure vessel would mean that the VVER-TOI should have a 120-year operational life. A 420-tonne ingot had been forged into one of these by OMZ-Spetsstal in December 2014. The new alloy was developed at Atomenergomash's Central Research Institute for Machine Building Technology (CNIITMASH). However the pressure vessel for Kursk II-1 delivered to the site by Atommash in September 2021 is made of nickel-free steel, to give a service lifetime of 100 years.

Petrozavodskmash makes steam generators and had the contract for RPV and various internals for the (now abandoned) Baltic 1 reactor. Izhorskiye Zavody was expected to supply these components for unit 2.

ZiO-Podolsk also makes steam generators, including those for Belene/Kozloduy 7.

Turbine generators for the new plants are mainly from Power Machines (Silovye Mashiny – Silmash) subsidiary LMZ.

Turbine Technology AAEM is a joint venture of Atomenergomash and GE producing low-speed turbines based on Alstom's Arabelle design, sized from 1200 to 1800 MWe, with GE generators. It produces the Arabelle units at AEM's Atommash plant at Volgodonsk, most recently for Akkuyu in Turkey and El Dabaa in Egypt.

Ukraine's Turboatom is offering a 1250 MWe low-speed turbine generator for the VVER-TOI. Rosenergoatom says it insists on

having at least two turbine vendors, and prefers three.

Reactor technology

Pressurized water reactors

VVER-1000, AES-92, AES-91

The main reactor design deployed until now has been the V-320 version of the VVER-1000 pressurized water reactor with 950-1000 MWe net output. It is from OKB Gidropress (Experimental Design Bureau Hydropress), has 30-year basic design life and dates from the 1980s. A later version of this for export is the V-392, with enhanced safety and seismic features, as the basis of the AES-92 power plant. All models have four coolant loops, with horizontal steam generators. Maximum burn-up is 60 GWd/tU. VVER stands for water-cooled, water-moderated energy reactor.

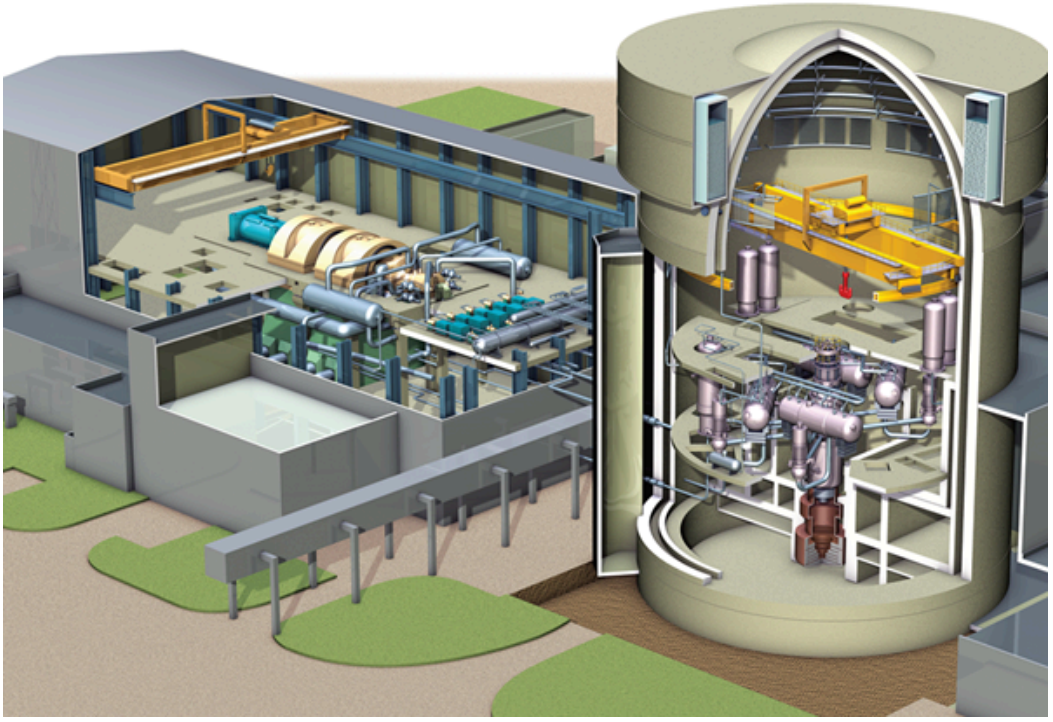
Advanced versions of this VVER-1000 with western instrument and control systems are being built at Tianwan in China and are being built at Kudankulam in India – as AES-91 and AES-92 nuclear power plants respectively. The former was bid for Finland in 2002 and for Sanmen and Yangjiang in China in 2005, while the AES-92 was accepted for the now-cancelled Belene project in Bulgaria in 2006. These have 40-year design life. (Major components of the two designs are the same except for slightly taller pressure vessel in AES-91, but cooling and safety systems differ. The AES-92 has greater passive safety features – 12 heat exchangers for passive decay heat removal; the AES-91 has extra seismic protection. The V-428 in the AES-91 is the first Russian reactor to have a core-catcher, V-412 in AES-92 also has core catcher.)

VVER-1200, AES-2006, MIR-1200

Development of a third-generation standardized VVER-1200 reactor of about 1170 MWe followed, as the basis of the AES-2006 power plant. Rosatom drew upon Gidropress, OKBM, Kurchatov Institute, Rosenergoatom, Atomstroyexport, three Atomenergoproekt outfits, VNIINPP and others. Two design streams emerged: one from Atomproekt in St Petersburg with V-491 reactor, and one from Atomenergoproekt in Moscow with V-392M reactor.

The V-491 provides about 1170 MWe gross, 1085 net, and the V-392M provides about 1199 MWe gross, 1114 net, both from 3200

MWt, along with about 300 MWt for district heating. This is an evolutionary development of the well-proven VVER-1000/V-320 and then the third-generation V-392 in the AES-92 plant (or the AES-91 for Atomproekt version), with longer operational lifetime (60 years for non-replaceable equipment, not 30), greater power, and greater thermal efficiency (34.8% net instead of 31.6%). Compared with the V-392, it has the same number of fuel assemblies (163) but a wider pressure vessel, slightly higher operating pressure and temperature (329°C outlet), and higher burn-up (up to 70 GWd/t). It retains four coolant loops. It can undertake daily load following down to 80% power. Refuelling cycle is up to 24 months. Core catchers filled with non-metallic materials are under the pressure vessel. Construction time for serial units is "no more than 54 months".



AES-2006 (Rosatom)

The lead units were built at Novovoronezh II (V-392M), starting operation in 2016, and at Leningrad II (V-491) in 2018. Both plants use Areva's (now Framatome) Teleperm safety instrument and control systems. Atomproekt's Leningrad II with V-491 reactor was quoted as the reference plant for the VVER units now operating at Tianwan in China. The two types of AES-2006 plant are very similar apart from the configuration of the safety systems. They are expected to run for 60 years with capacity factor of 92%, and with Silmash turbine generators. They have enhanced safety including that related to earthquakes and aircraft impact with some passive safety features and double containment. However, it appears that

only six will be built domestically – two V-392M and four V-491 – before moving onto the VVER-TOI, with potential for international design certification.

For the Novovoronezh V-392M units Atomernergoproekt Moscow has installed what it calls dry protection, a 144-tonne structure surrounding the reactor core that reduces emission of radiation and heat. It consists of a steel cylinder with double walls, 7m diameter, with the space between them filled with specially formulated concrete. This gives it better aircraft crash resistance than V-491. They have passive decay heat removal by air circulation.

Atomproekt's Leningrad V-491 units have four trains of active safety systems, with water tanks high up in the structure to provide water cooling for decay heat, and is more suited to Finland and central Europe rather than seismic sites (DBGM is only 250 Gal). Atomproekt's AES-2006 has two steam turbine variants: Russian Silmash high-speed version for Russia, or Alstom Arabelle low-speed turbine as was proposed for Hanhikivi and MIR-1200.

For Europe, the basic Atomproekt V-491 St Petersburg version has been slightly modified by Atomproekt as the **MIR-1200** (Modernized International Reactor), and bid for Temelin 3&4. It was also selected for Hanhikivi in Finland, as AES-2006E, with "extended list of accidents and external impacts" including higher seismic tolerance.

Atomstroyexport differentiates the VVER-1200M at Novovoronezh and for Rooppur from the VVER-1200E in Belarus. The differences are in the structure and layout of the safety systems.

VVER-1300, VVER-TOI

A further evolution, or finessing, of Moscow Atomenergoproekt's version of the AES-2006 power plant with the V-392M reactor is the VVER-TOI (typical optimized, with enhanced information) design for the AES-2010 plant, the VVER-1300 reactor being designated V-510 by Gidropress. Rosatom says that this is planned to be standard for new projects in Russia and worldwide, with minor variations (such as the cheaper VVER-1300A). It has an upgraded pressure vessel with four welds rather than six, and will use a new steel which "removes nearly all limitations on RPV operation in terms of radiation embrittlement of metal", making possible a service life of more than 60 years with 70 GWd/t fuel burn-up and 18 to 24-month fuel cycle. It has increased power to 3312 MWt, 1255 MWe gross (nominally 1300), improved core design still with 163 fuel assemblies to increase cooling reliability, larger steam generators, further development of passive safety

with at least 72-hour grace period requiring no operator intervention after shutdown. It targets lower construction and operating costs, and 40-month construction time. It is claimed to require only 130-135 tonnes of natural uranium (compared with typical 190 tU now) per gigawatt year. It will use a low-speed turbine-generator. It can undertake daily load following down to 50% thermal power, ramping up at 1%/minute and down at 3%/minute, and has significant frequency control capability compared with AES-2006 reactors.

The project was initiated in 2009 and the completed design was presented to the customer, Rosenergoatom at the end of 2012. The design aim was to try and save 20% of the cost. It was submitted to Rostechnadzor in 2013 for licensing, and was certified by the European Utility Requirements for LWR Nuclear Power Plants (EUR) organization in 2019. EUR approval is seen as basic in many markets, notably China.

The first units are under construction at Kursk. In June 2012 Rosatom said it would apply for VVER-1200 design certification in UK and USA, through Rusatom Overseas, with the VVER-TOI version, but this did not proceed. Development involved OKB Gidropress (chief designer), NRC Kurchatov Institute (scientific supervisor), All-Russian Scientific and Research Institute for Nuclear Power Plant Operation (VNIIAES – architect-engineer), and NIAEP-ASE jointly with Alstom (turbine island designer). V-509 and V-513 reactors are variants of V-510, the V-513 as VVER-1300.

VVER-1300A

This is presented as a cheaper variant of the VVER-TOI/V-510 reactor, and has two new PGV-1300A steam generators, decreased metal content and decreased containment diameter. It is proposed as a variant of the VVER-TOI.

VVER-600

Since 2008 OKB Gidropress with SPb AEP and Kurchatov Institute has also been developing a two-loop VVER-600 (project V-498) from V-491 (1200 MWe, four-loop), using the same basic equipment but no core catcher (corium retained in RPV), as a Generation III+ type. In December 2011 Gidropress signed a contract with the Design and Engineering Branch of Rosenergoatom for R&D related to the VVER-600 reactor, though this was not then part of any federal Rosatom program. Gidropress presented the design to Rosenergoatom in February 2013, saying a project package could be ready in two years. It will be capable of load-following, and would have a 60-year life. Rosenergoatom has

been considering it for the Baltic plant site as a straightforward option to replace the 1200 MWe units, and it is now planned for Kola. It has high export potential, and Gidropress, NIAEP and Kurchatov have been progressing it slowly. The VVER-600 supercedes the VVER-640 in Gidropress plans,* and Rosatom envisages its deployment in 2020s.

* The **VVER-640** (V-407), an 1800 MWt, 640 MWe design originally developed by Gidropress jointly with Siemens. It had advanced safety features (passive safety systems). After apparently beginning construction of the first at Sosnovy Bor, funds ran out and it disappeared from plans. However, it came back on the drawing boards, now as a Generation III+ type, with four cooling loops, low power density, low-enriched fuel (3.6%), passive safety systems, 33.6% thermal efficiency and only 45 GWd/t burn-up. In March 2013 SPbAEP (merged with VNIPIET to become Atomproekt) said that subject to Rosatom approval it could have a VVER-640 project ready to go possibly at the Kola site by the end of 2014. The project partners – Atomenergomash, OKB Gidropress, Central Design Bureau for Marine Engineering (CDBME) of the Russian Shipbuilding Agency, OMZ's Izhorskiye Zavody, Kurchatov Institute, and VNIPIET – "confirmed its readiness for updating aiming at commercialization." In May 2013 Atomenergoproekt said it has already been discussing with VNIPIET the feasibility and practicability of using the VVER-640 project "as the starting point for the development of next-generation medium-power nuclear power plants, including with the use of passive safety systems."

VVER-1500

About 2005 Rosatom (the Federal Atomic Energy Agency) promoted the basic design for VVER-1500 pressurized water reactors by Gidropress as a priority. Design was expected to be complete in 2007, but the project was shelved in 2006. It was a four-loop design, 42,350 MWt producing 1500 MWe gross, with increased pressure vessel diameter to 5 metres, 241 fuel assemblies in core enriched to 4.4%, burn-up up 45-55 and up to 60 GWd/t and life of 60 years.

VVER-1800

This is a development of the VVER-1300A, but with three loops, using steam generators and circulation pumps from it, and reactor pressure vessel and internals from VVER-1500, which it supercedes. However, development is paused.

VVER-SKD-1700

A Generation IV Gidropress project in collaboration with the Generation IV International Forum is the supercritical VVER (VVER-SKD or VVER-SCWR) with higher thermodynamic efficiency (45%) and higher breeding ratio (0.95) and oriented towards the closed fuel cycle. Focus is on structural materials and fuels. The main version is 3830 MWt, 1700 MWe, with 540°C operating temperature. The SPA Central Research Institute of Machine Engineering Technology (TsNIITMASH) in Moscow and OKB Giidropress are involved in the draft proposals. OKB Gidropress says that "Such reactors are expected to increase significantly thermal energy conversion efficiency, move to the fast neutron spectrum in the reactor core and, by thus, substantially improve parameters of breeding of the secondary nuclear fuel in the reactor." Also referred to as VVER-1700, V-393. Rosatom is reported to be developing it to a full design and bidding to build a prototype ahead of other SCWR designs in Europe, Canada, China and Japan.

A **PSKD-600** fast reactor version (1430 MWt, 600 MWe) is also being developed, with primary circuit temperature of 500°C, hence also a secondary steam circuit, and breeding ratio >1.

VVER-I

The VVER-I is a range of small modular reactors under development, with integral steam generators. It is proposed to be built at a central factory and then transported by ship, rail and road to the construction site with a high degree of modularization. There are three planned sizes of 100, 200 and 300 MWe (VVER-I-200, etc).

VVER with spectral core

As a step in the direction of breeder reactors and to obtain better fuel utilization, Gidropress has a 'Super' VVER, the VVER-S (or VVER-SM) project with spectral core control, changing the neutron spectrum by having removable diplacers (e.g. 13) in each fuel assembly (in 132 fuel assemblies out of 241 in a 3300 MWt reactor). It is developed as a successor to the VVER-TOI after 2030, and will be able to operate efficiently using 100% MOX fuel. Spectral regulation is achieved by changing the water-uranium ratio in the process of fuel burn-up. Compensating for burn-up using burnable poisons is minimized and is replaced by the shift of neutron spectrum. Fresh fuel has the so-called displacers in each assembly to decrease the moderator (water) quantity in the core and making more of a fast neutron spectrum, which decreases the reactivity coefficient and increases the breeding ratio of nuclear fuel

(in contrast to the regulation with boron in the water or burnable poisons).

As burn-up proceeds and fission products build up, reducing the breeding ratio, the displacers are gradually extracted, which increases the water-uranium ratio and slows the neutron spectrum to near thermal. This makes it possible to use the Pu-239 and Pu-241 isotopes, effectively accumulated in the 'rigid' spectrum from the U-238. Water-uranium ratio in the core is about 1.5 at start of cycle and about 2.0 at end.

Small VVERs

After many years of promoting the idea, in 2006 Rosatom approved construction of a nuclear power plant on a barge (floating power module, FPM) to supply power and heat to isolated coastal towns. See Floating nuclear power plants subsection above. Building such plants as small modular reactors (SMRs) on land has also been considered.

Twin OK-900A reactors from OKBM Afrikantov were fitted to six icebreakers from 1975. Derived from these, a KLT-40 reactor was fitted to the *Sevmorput* cargo ship in 1988, and single KLT-40M units were fitted to two shallow-draught icebreakers in 1989-90.

Two OKBM Afrikantov **KLT-40S** reactors derived from the OK-900A reactors in icebreakers, but with low-enriched fuel (less than 20% U-235), supply 70 MWe of power plus 586 GJ/hr (5.1 PJ/yr) of heat. They are mounted on a 21,500 tonne, 144 m long, 30 m wide barge. Refuelling interval is 3-4 years onsite, and at the end of a 12-year operating cycle the whole plant is returned to a shipyard (Zvezdochka, near Sevmash has been mentioned) for a two-year overhaul and storage of used fuel, before being returned to service. Each reactor is 140-150 MWt and can deliver 38.5 MWe if no cogeneration is required.

The smaller **ABV** reactor units were under development by OKBM Afrikantov, with a range of sizes from 45 MWt (ABV-6M) down to 18 MWt (ABV-3), giving 4-18 MWe outputs. The VVER units are compact, with integral steam generator. The whole unit of some 200 tonnes (ABV-6) would be factory-produced for ground or barge mounting. A single ABV-6M would require a 3500 tonne barge; the ABV-3, 1600 tonne. The core is similar to that of the KLT-40 except that enrichment is 16.5% and average burn-up 95 GWd/t. It has natural circulation in the primary circuit. Refuelling interval is about 8-10 years, and service life about 50 years. In mainly desalination mode the ABV-6M is expected to produce 55,000 m^3/day of potable water by reverse osmosis. The company said at the end of 2009

that an ABV-R7D would cost RUR 1.5 billion, but that Rosatom preferred the larger and proven KLT-40 design.

OKBM Afrikantov has developed a new compact icebreaker reactor – **RITM-200** – to replace the current 171 MWt KLT 40M and OK-900A reactors. This is an integral 175 MWt, 50-55 MWe PWR with inherent safety features. Two of these, as in the new LK-60 icebreakers, will give 60 MW shaft power via twin turbine generators and three motors. At a 65% capacity factor, fuel reloading is required after seven years and major overhaul period is 20 years. Its service lifetime is up to 60 years. The reactor plant mass is 2200 tonnes for two. Fuel enrichment is almost 20%. TVEL started making the fuel in 2016, with 4.5 TWh from each load. Reactor containment dimensions are 6 x 13.2 x 15.5 m for two.

The **RITM-200B** is a 209 MWt version for use as a single unit in smaller icebreakers.

The **RITM-400** of 315 MWt each, with twin units to deliver 120 MW propulsion through four electric motors is being developed for larger icebreakers. The reactor plant mass will be 1936 tonnes each. Operating lifetime is 40 years and fuel cycle 5 years. Reactor containment dimensions are 9 x 8.2 x 17.5 m. The first is due to be commissioned in 2027.

For optimized floating nuclear power plants (OFPUs), two **RITM-200M** reactors replace twin KLT-40S reactors and require only a 12,000 tonne barge, much smaller than for the KLT-40S. The reactor mass of twin RITM-200M units is only 2600 tonnes, compared with 3740 t for two KLT-40S reactors. They are 175 MWt/50 MWe each and refuelling is every 10 years. Reactor containment dimensions are 6.8 x 14.6 x 16 m for two units. Rosatom is planning three such units at Cape Nagloynyn in Chakotka to supply 330 MWe to the Baimskaya copper mining project south of Bilibino and Pevek from 2027, with a fourth in reserve. Construction of the first barge commenced with a keel-laying ceremony in August 2022 at Wison Heavy Industries in Nantong, China.

In December 2022 Rosatom announced that the development of nuclear fuel for its OFPUs, including RITM-200S reactors, had been completed.

For land-based plants the State Specialized Design Institute (SSDI or GSPI) finalized the design of a single-unit **RITM-200N** plant in September 2018. This has 190 MWt power (55 MWe), a fuel cycle of 5 or 6 years and a service lifetime of 60 years. Reactor containment dimensions are 6 x 6 x 15.5 m. Rosatom signed an

electricity supply agreement in 2020 and then a development agreement in September 2021 with the government of Sakha (Yakutia) for the first land-based small modular reactor at Ust-Kuyga. Rostechnadzor licensed this RITM-200N in August 2021. A construction licence was granted in April 2023. It will replace coal and diesel capacity in the Ust-Yansky district and also supply the Kyuchus gold mine project in the Verkoyansky district. It is expected to halve the local cost of electricity.

Exports of combined power and desalination units is planned, with China, Indonesia, Malaysia, Algeria, Cape Verde and Argentina being mentioned as potential buyers, though Russia would probably retain ownership of the plant with operational responsibility, and simply sell the output. Rosatom has formed a group of expert desalination advisors as part of a strategy to sell its thermal desalination technologies. It is targeting world regions where clean water is scarce as part of its drive for leadership in the global nuclear market.

VBER-300, VBER-200 to 500

OKBM Afrikantov's **VBER-300** PWR is a 325 MWe gross, 295 MWe net, PWR unit developed from naval power reactors and was originally envisaged in pairs as a floating nuclear power plant.* As a cogeneration plant it is rated at 200 MWe and 1900 GJ/hr for heat or desalination. The reactor is designed for a 60-year operating lifetime and 90% capacity factor. It was planned to develop it as a land-based unit with Kazatomprom, with a view to exports, and the first unit was to be built at Aktau in Kazakhstan. However, this agreement stalled, and OKBM has been looking for a new partner to develop it. Two demonstration units were proposed at Zheleznogorsk for the Mining & Chemical Combine (MCC), costing some $2 billion. MCC preferred the VBER design to the VK-300. There is also discussion of transportable nuclear power units with a VBER-300 reactor set up on a stationary ice-resistant platform for northern LNG production, with one plant providing for about 5 Mt LNG per year.

* Twin VBER-300 units would use a 49,000 tonne barge 170 m long and 62 m wide.

In October 2012 a VBER-500 design was announced by OKBM Afrikantov, with design to be completed in about 2015 in collaboration with NIAEP. In fact OKBM offers 200 to 600 MWe plants “based on the standard 100 MWe module.” They are based on over 6000 reactor-years of experience with naval reactors. The VBERs were not part of any federal program, but the VBER-500

has explicit support from Rosenergoatom, with Kola replacement in view, and also perhaps as an alternative for the Baltic plant's aborted 1200 MWe units.

AST-500

The AST-500 was an early design primarily for district heating. The integral PWR was designed by OKBM and built by Atommash, with the first one installed at Gorky and ready to start up in September 1989, but never operated. It shares some design features with the VK-300.

Russian PWR nuclear power reactors*

Generic reactor type	Reactor plant model	Whole power plant
VBER-300		(under development) OKBM, 325 MWe gross, based on KLT-40
VVER-210	V-1	prototype VVER, Novovoronezh 1
VVER-365	V-3M	Novovoronezh 2
VVER-440	V-179	Novovoronezh 3-4, prototype VVER-440
	V-230	Kola 1-2, EU units closed down
	V-270	Armenia 1-2, based on V-230
	V-213	Kola 3-4, Rovno 1-2, Loviisa, Paks, Dukovany, Bohunice V2, Mochovce
	V-318	Cuba, based on V-213, full containment & ECCS
VVER-640	V-407	(under development), Gen III+, Gidropress
VVER-300	V-478	(under development, based on V-407), Gen III+, Gidropress
VVER-600	V-498	(under development by Gidropress, based on V-491), Gen III+, proposed for Kola
VVER-1000	V-187	Novovoronezh 5, prototype VVER-1000
	V-302	South Ukraine 1
	V-320	most Russian & Ukraine plants, Kozloduy 5-6, Temelin 1-2
	V-338	Kalinin 1-2, South Ukraine 2
	V-446	based on V-392, adapted to previous Siemens work, Bushehr 1
	V-413	AES-91
	V-428	AES-91 Tianwan and Vietnam proposal, based on V-392, Gen III
	V-428M	Tianwan 4&5, later version

Generic reactor type	Reactor plant model	Whole power plant
	V-412	AES-92 Kudankulam, based on V-392, Gen III
	V-392	AES-92 – meets EUR standards, Armenia, Khmelnitsky 3-4, Gen III
	V-392B	AES-92
	V-466	AES-91/99 Olkiluoto bid, also Sanmen, developed from V-428, Gen III
	V-466B	AES-92 Belene/Kozloduy 7, Jordan?, developed from V-412 & V-466, 60-year lifetime, 1060 MWe gross, Gen III, Gidropress
	V-528	Bushehr 2&3 version of V-466B
VVER-1200	V-392M	AES-2006 by Moscow AEP and Gidropress, Novovoronezh; Developed from V-392 and V-412, Gen III+, 1170 MWe gross, more passive safety than V-491, developed to VVER-TOI
	V-491	AES-2006 Leningrad, Baltic, Belarus, Tianwan 7&8, Ninh Thuan 1 bid; developed from AES-91 V-428 by Atomproekt and Gidropress, Gen III+, 1170 MWe gross, developed to MIR-1200 for EUR
	V-508	MIR-1200 from V-491 for EUR, Temelin bid
	V-509	Akkuyu version, based on Novovoronezh V-392M
	V-522	Hanhikivi version of V-491 AES-2006E
	V-523	Rooppur version of V-392M, Novovoronezh reference, AES-2006M
	V-527	Paks II version of V-491 AES-2006E
	V-529	El Dabaa version of V-491 AES-2006E
VVER-1200A	V-501	Concept proposal AES-2006, but two-loop, shelved in 2011
VVER-1300	V-488	AES-2006M, developmental model, Gen III+, Gidropress
	V-510	AES-2010, Generation III+ VVER-TOI, 1250 MWe gross, developed by Moscow AEP from V-392M, Nizhny Novgorod, Kursk II, Smolensk II, Central, Tatar
	V-513	Upgraded V-392M, VVER-TOI
VVER-1300A	?	Cheaper variant of VVER-TOI
VVER-1500	V-448	Gidropress, Gen III+, shelved in 2006

Generic reactor type	Reactor plant model	Whole power plant
VVER-1800	?	(concept proposal) three loops, based on 1300A and 1500
VVER-SCP	V-393	being developed, Supercritical, Gen IV

AES=NPP. Early V numbers referred to models which were widely built in several countries, eg V-230, V-320. Then the V-392 seemed to be a general export version of the V-320. Later V numbers are fairly project-specific. Broadly the first digit of the number is the VVER generation, the second is the reactor system and the third – and any suffix – relates to the building. Generation III or III+ ratings are as advised by Gidropress, but not necessarily accepted internationally.

* V-392M has two active safety channels, while V-491 has four, and turbine hall layouts are also different. In the V-392M there is a focus placed on avoidance of redundancy, aiming at higher cost-effectiveness of the plant construction and operation. Both V-392M and V-491 designs include a common emergency core cooling system (ECCS) passive section, but in the V-392M the ECCS active section is represented by a combined two-channel high and low pressure system, while the V-491 utilizes a segregated four-channel high and low pressure system. The V-392M design features a closed two-channel steam generator emergency cool-down system, whereas the V491 uses a traditional four-channel emergency feedwater system. To mitigate consequences of beyond design basis accidents involving total loss of AC power sources, both designs use a passive heat removal system, which is air-cooled in the V-392M and water-cooled in the V-491. Additionally, the V-392M design is fitted with a four-channel emergency passive core flooding system.

While Gidropress is responsible for the actual 1200 MWe reactor, Moscow AEP and Atomproekt St Petersburg are going different ways on the cooling systems, and one or the other may be chosen for future plants once Leningrad II and Novovoronezh II are operating. Passive safety systems prevail in Moscow's V-392M design, while St Petersburg's V-491 design focuses on active safety systems based on Tianwan V-428 design.

VK-300 BWR

The **VK-300** boiling water reactor is being developed by the Research & Development Institute of Power Engineering (NIKIET) for both power (250 MWe) and cogeneration with desalination (150 MWe plus 1675 GJ/h). It has evolved from the Melekess VK-50

BWR at Dimitrovgrad, but with standard components used wherever possible, *e.g.* the reactor vessel of the V-320. A feasibility study on building four cogeneration VK-300 units at Archangelsk was favourable, each pair conservatively delivering 3 TWh and 16 PJ/yr heat, but this has not proceeded. NIKIET is still promoting the design but the VBER-300 seems a stronger prospect.

RBMK/LWGR

A development of the RBMK light water graphite reactor was the MKER-800, with much improved safety systems and containment, but this too has been shelved. Like the RBMK itself, it was designed by VNIPIET (All-Russia Science Research and Design Institute of Power Engineering Technology) at St Petersburg.

High temperature gas-cooled reactors

In the 1970-80s OKBM undertook substantial research on high temperature gas-cooled reactors (HTRs). In the 1990s it took a lead role in the international **GT-MHR** (Gas Turbine-Modular Helium Reactor) project based on a General Atomics (US) design. Preliminary design was completed in 2001 and the prototype was to be constructed at Seversk (Tomsk-7, Siberian Chemical Combine) by 2010, with construction of the first four-module power plant (4x285 MWe) by 2015. Initially it was to be used to burn pure ex-weapons plutonium, and replace production reactors which supplied electricity there to 2010. However, the project lapsed.

In the longer-term perspective, HTRs were seen as important for burning actinides, and later for hydrogen production. OKBM is now in charge of Russian HTR collaboration with China as interest in HTRs is revived. In 2019 Rosatom outlined its HTR program, with the focus on hydrogen production by adiabatic conversion of methane with utilization of CO_2. Thermochemical hydrogen production from water is also envisaged.

In 2015 Rosatom agreed with Indonesia's BATAN for the pre-project phase of construction of an experimental multi-functional HTR there.

Fast reactors

For context, see also above section on Transition to Fast Reactors.

BN-600 Beloyarsk 3

The BN-600 fast neutron reactor operating since 1980 is licensed to 2040. It is a three-loop pool type reactor of 1470 MWt,

600 MWe gross and 560 MWe net. Due to progressive modification, its fuel burn-up has increased from 7% (design value) to 11.4%. It provides heat for Zarechny town as well as electricity from three 200 MWe turbine generators.

BN-800 Beloyarsk 4

The Beloyarsk 4 BN-800 fast reactor designed by OKBM Afrikantov was intended to supersede the BN-600 unit 3 at Beloyarsk, and utilize MOX fuel with both reactor-grade and weapons plutonium. The RUR 146 billion project was delayed by lack of funds following construction start in 2006. Atomproekt was the general designer. It was initially represented as the first Generation III reactor which, after 2020, was to take a large share of Russian capacity as older designs were phased out.

This first (and probably only Russian) BN-800 unit first started up in June 2014, with first power to the turbine in November 2015. It was grid connected in December 2015 and reached full power in August 2016. It is 2100 MWt, 864 MWe gross, 789 MWe net, with fuel burn-up of 66 GWd/t initially, increasing to 100 GWd/t. It is essentially a demonstration unit for fuel and design features for the BN-1200, or as Rosatom said in September 2015: the BN-800 has been created for testing elements of closing the nuclear fuel cycle rather than electricity generation, though it did produce 13.7 TWh in its first 36 months operation. Uralenergostroy is the general civil contractor for both Beloyarsk reactors, and sees the BN-800 as a bridge to significantly different future designs such as the BN-1200.

Beloyarsk 4 (Rosatom)

Rosenergoatom said: "For us, the BN-800 is not only the basis for development of a closed nuclear fuel cycle. It is also a test case for technical solutions that will later be used for commercial production of the BN-1200. Among other things, the BN-800 must answer questions about the economic viability of potential fast reactors ... if such a unit has more functions than to generate electricity, then it becomes economically attractive. That's what we have to find out." The Beloyarsk plant director said: "The main objective of the BN-800 is [to provide] operating experience and technological solutions that will be applied to the BN-1200."

After first criticality in June 2014 there was a long preparatory period with a year's delay before return to criticality in mid-2015 due to a technical issue with the fuel assemblies, which had to be redesigned when it became clear that not enough MOX fuel would be available for an initial core load.

In January 2020 the reactor was loaded with mixed oxide (MOX) fuel incorporating recycled plutonium. In September 2022 the reactor became the first power reactor to operate on a full MOX core, and has since operated reliably with this fuel configuration. The assemblies were produced at Zheleznogorsk, with plutonium recycled from conventional fuel, mixed with depleted uranium left over from enrichment plants. The BN-800 will require 1.84 tonnes each year of reactor-grade plutonium recovered from 190 tonnes of used fuel from conventional VVER reactors.

The unit does not have a breeding blanket and breeding ratio is quoted as 1.0, though the version designed for Sanming in China has up to 198 DU fuel elements in a blanket. Further reactor details in the information page on Fast Neutron Reactors

In May 2009 St Petersburg Atomenergopoekt (SPb AEP, now Atomproekt) said it was starting design work on a BN-800 reactor for China, where two were planned at Sanming – Chinese Demonstration Fast Reactors (CDFRs). They would use pelletized MOX fuel, initially from MCC. A high-level agreement was signed in October 2009, then another in November 2012, and an intergovernmental agreement relating to them was expected in 2012, but the project did not proceed.

BN-1200 Beloyarsk 5

The BN-1200 reactor is being developed by OKBM Afrikantov in Zarechny, partly funded by federal nuclear technology programme (FTP2010). The BN-1200 will produce 2900 MWt (1220 MWe gross), has a 60-year design operating lifetime, simplified refuelling,

and burn-up of up to 120 GWd/t, with breeding ratio quoted as 1.2 to 1.4, using oxide or nitride fuels.

OKBM said that the capital cost would be the same as for the VVER-TOI. Rosatom's Scientific & Technical Board reviewed it along with cost estimates in August 2015. OKBM Afrikantov announced completion of the detailed design in May 2017. SPb AEP (merged with VNIPIET to become Atomproekt) is the general designer. Rosatom sees this as a "Generation IV design with natural security" – an element of the *Proryv* (Breakthrough) project*, with closed fuel cycle. Further reactor details in the information page on Fast Neutron Reactors.

* for large fast reactors, BN series and BREST.

OKBM earlier expected the first BN-1200 unit with MOX fuel to be commissioned in 2020, then eight more to 2030, moving to dense nitride U-Pu fuel with closed fuel cycle.

In May 2012 Rosenergoatom started environmental assessment for a BN-1200 unit as Beloyarsk 5, with an evaporative cooling tower. In April 2015 Rosenergoatom said that construction decision would be delayed to at least 2020, as it wanted to improve the fuel and review the economic viability of the project. Federal financing and Rosatom funds of RUR 102 billion ($3.3 billion) were envisaged. Then in August 2019 it was decided to delay construction of the BN-1200 to the 2030s due to a combination of factors including the economic situation, the reduced projected demand for electric power, uncertainty about the economic comparison with the VVER-1200, and the delay in proving technology for the production of dense nitride fuel. In April 2023 Russia's JSC Institute Orgenergostroy commenced the engineering survey work for the planned unit. In April 2025 Rostechnadzor issued a construction licence for the unit. Civil construction work began in July 2025, with first nuclear concrete planned for 2027 and completion targeted for 2034.

The Chelyabinsk regional government has planned for up to three BN-1200 units to be built at the proposed South Urals plant. A draft energy plan in 2024 specified two units.

A BN-1800 was briefly under development.

BREST-300 Seversk

The BREST-300 lead-cooled fast reactor (bystry reaktor so svintsovym teplonositelem) is another innovation, from NIKIET, with the first unit earlier being proposed for Beloyarsk 5. This is a new-

generation fast reactor which dispenses with the fertile blanket around the core and supersedes the BN designs to give enhanced proliferation resistance. Lead cooling enables greater utilization of minor actinides than in BN reactors. In February 2010 a government decree approved RUR 40 billion ($1.3 billion) funding for an initial 300 MWe BREST unit (at the Siberian Chemical Combine in Seversk rather than Beloyarsk).

In September 2012 Rosatom announced that a pilot demonstration BREST-300 fast reactor with associated fuel cycle facilities including mixed uranium-plutonium nitride (MNUP) fuel fabrication would be built at the Siberian Chemical Combine (SCC). A construction schedule was presented at a *Proryv* (Breakthrough) project meeting at SCC in March 2013. The State Environmental Commission of the Federal Service for Supervision of Natural Resources (Rosprirodnadzor) issued a positive statement on the construction licence application package for the pilot demonstration power complex (PDPC) and fuel fabrication module in June 2014, and Rostechnadzor issued a licence in 2014 for the fuel module which is due in operation in 2023. The MNUP fuel is derived from recycled plutonium and depleted uranium. The fuel has been extensively tested in the BN-600 reactor at Beloyarsk since 2015, and by the end of 2020 more than 1000 MNUP fuel assemblies had been produced in SCC's pilot plant.

NIKIET finished the BREST design in 2014, and the State Expert Review Authority (Glavgosekspertiza) approval of the design documentation was reported in December 2018. Working documentation for preparation of the site and construction of the reactor was to be prepared in 2016, along with a preliminary report on the safety aspects of the project. Atomproekt is the general designer. A government decree in August 2016 ordered construction by 2025, but in October 2018 Rosatom announced that the reactor would not begin commercial operation before 2026. It is now not expected to begin operation before 2028-2029.

Earlier, in November 2019 SCC awarded a contract to Titan-2 to build the first unit. In February 2021 Rostechnadzor issued a construction licence to SCC Seversk for the BREST-OD-300 reactor and construction started in June 2021. The reactor pressure vessel is reinforced concrete and it operates at near atmospheric pressure. The fuel fabrication/refabrication module at the PDPC entered pilot commercial operation in December 2024 and began producing prototype fuel assemblies with depleted uranium nitride fuel pellets. The main turbine and generator were delivered to the reactor site in August 2025.

If BREST is successful as a 300 MWe unit, a 1200 MWe (2800 MWt) version (BREST-1200) will follow. The PDPC comprises three phases: the mixed uranium-plutonium nitride fuel fabrication/refabrication module; a nuclear power plant with BREST-OD-300 reactor; and a used nuclear fuel reprocessing module. BREST-300 has 17.6 tonnes of fuel, BREST-1200 about 60 tonnes.

SVBR-100 Dimitrovgrad

The **SVBR-100** (Svintsovo-Vismutovyi Bystryi Reaktor – lead-bismuth fast reactor) is a modular lead-bismuth cooled fast neutron reactor designed by OKB Gidropress in Podolsk and c. It is a small modular reactor (SMR) so that larger power plants can be built incrementally and comprise several 100 MWe modules. It was intended to meet regional needs in Russia and abroad. The design is also known as the MTBF-100.

The pilot 101 MWe SVBR-100 unit was planned to be built next to RIIAR Dimitrovgrad by AKME-engineering – a 50-50 joint venture of Rosatom and private company En+ Group set up to develop and operate this. In 2010 AKME Engineering contracted with Atomenergoproekt to design the pilot SVBR-100 modular fast reactor, with the State Scientific Centre of the Russian Federation – Institute for Physics and Power Engineering (IPPE) at Obninsk. RUR 13.23 billion was allocated for this in February 2010, including RUR 3.75 billion from the federal budget. However, in December 2014 Rosatom said that the cost had escalated to RUR 36 billion, more than twice the original estimate, making it “less commercially attractive.” This figure remained current at the end of 2016. In October 2015 Rosatom reported that "experts have confirmed there are no scientific or technical issues that would prevent completion of the project and obtaining a construction licence." In November 2016 Rosatom said it expected to work out the main specifications for construction of the SVBR-100 by mid-2017, but in 2018 the project had been dropped.

Each 100 MWe fast reactor module with lead-bismuth primary coolant would be 4.5 x 8.2 metres, built in factories and delivered to site. The 280 MWt reactor has integral design and forced convection circulation of primary coolant at up to 500°C with two main circulation pumps but with passive cooling after shutdown. Fuel would be low-enriched (16.5%) uranium or MOX initially, later possibly nitride. Refuelling interval would be 7-8 years and there is no breeding blanket. Design lifetime was 60 years. It was proposed as a replacement for Novovoronezh 3&4 (in the present reactor halls), and for Kozloduy in Bulgaria. It is described by Gidropress as

a multi-function reactor, for power, heat or desalination, to meet regional needs in Russia and abroad. Serial production was envisaged from 2024.

This was a pilot project for Russia in terms of the nuclear industry in large-scale high-tech projects since it sought the participation of private capital by mid-2017. Rosatom looked for additional investors in the project to enable it to proceed, but despite the new cooperation agreement with China in November 2016 including fast breeder reactors the project lapsed. See also Small Nuclear Reactors information page.

Another new reactor, also described as a multi-function fast reactor – **MBIR** – is being built at the Research Institute of Atomic Reactors (RIAR) at Dimitrovgrad. See above section on Transition to Fast Reactors and the R&D section in the information page on Russia's Nuclear Fuel Cycle (as this is not essentially a power reactor).

International collaboration

From 2001 Russia has been a lead country in the IAEA Project on Innovative Nuclear Reactors and Fuel Cycles (INPRO). In 2006 Russia joined the Generation IV International Forum, for which the OECD Nuclear Energy Agency (NEA) provides the secretariat. Russia is also a member of the NEA's Multinational Design Evaluation Programme which is increasingly important in rationalising reactor design criteria.

Export of nuclear reactors

The Ministry of Foreign Affairs is responsible for promoting Russian nuclear technologies abroad, including building up a system of Rosatom foreign representatives in Russian embassies. This is backed up by provision of substantial competitive finance for nuclear construction in client countries, as well as readiness to take equity or even build-own-operate (BOO) as in Turkey. Early in 2016 Rosatom said that Russia's GDP gained two roubles for every one rouble invested in building nuclear power plants abroad, as well as enhanced trade.

As of 2024, Rosatom's package of foreign orders was approximately $200 billion.

The company sees its strength as its ability to make an integrated offer for its nuclear power plants, offering not only turnkey

construction and fuel, but also training, services, infrastructure development, legal and regulatory structures, etc. in a single package. Rosatom said in November 2015 that due to its integrated structure, the cost (LCOE, levelized cost of energy) of new VVER reactors is no more than $50-$60 per MWh in most countries.

In 2016 Rosatom and the Bank for Development and Foreign Economic Affairs (Vnesheconombank) agreed to develop their cooperation to support Rosatom's investments in projects overseas. The agreement reflects the bank's "new strategic priorities". Rosatom said that “implementation of projects in the framework of the signed agreement will help to address the global challenges of the nuclear industry and increase the energy security of the Russian Federation." It will "contribute to the growth of the Russian economy and the expansion of Russia's presence in the global nuclear energy market," Rosatom said.

Atomstroyexport (ASE) has largely completed three reactor construction projects abroad, all involving VVER-1000 units. First, it took over building a reactor for Iran at the Bushehr power plant, a project commenced by Siemens KWU but then aborted. That reactor is now operating, and ASE is building a second unit at the site. Then it sold two AES-91 units to China for Jiangsu Tianwan at Lianyungang (both now operating) and two AES-92 units to India for Kudankulam (both now operating).

Russia is currently constructing about 20 units abroad (see below). A variety of financing arrangements are being used. China and Iran are paying for plants directly, India is benefitting from substantial Russian finance, whilst Bangladesh and Egypt are relying on major loans. A build-own-operate model using Russian finance but with guaranteed long-term electricity price is being pioneered in Turkey. A planned plant in Finland was to involve 34% Russian equity, but that has now been cancelled.

Export sales and prospects for Russian nuclear power plants (post-Soviet)

Country	Plant	Type	Status
Iran	Bushehr 1	VVER-1000/V-446	operable
China	Tianwan 1&2	AES-91	operable
	Tianwan 3&4	AES-91	operable

Country	Plant	Type	Status
India	Kudankulam 1&2	AES-92	operable
Belarus	Ostrovets 1&2	AES-2006/V-491	operable, loan for 25 years to finance 90% of the contract.
Operable: 9			

Country	Plant	Type	Status, financing
India	Kudankulam 3-6	AES-92	Construction start June 2017 and Oct 2017, and June 2021 and December 2021
Bangladesh	Rooppur 1&2	VVER-1200/V-392M	Construction start Nov 2017 and July 2018, loan organized for 90%
Turkey	Akkuyu 1-4	VVER-1200/V-509	Construction start April 2018, April 2020, March 2021, July 2022, BOO
Iran	Bushehr 2	AES-92/V-466B	Construction start Nov 2019. Phase II 18% complete. Construction suspended March 2026 due to military conflict.
China	Tianwan 7&8	VVER-1200/V-491	Construction start May 2021 and February 2022. Unit 7 hot tests completed December 2025
China	Xudabao 3&4	VVER-1200/V-491	Construction start May 2021 and May 2022
Egypt	El Dabaa 1-4	VVER-1200/V-529	Construction start July 2020, Nov 2020, May 2023, Jan 2024
Hungary	Paks II-1	VVER-1200/V-529	Construction start Feb 2026
Construction: 20			

Country	Plant	Type	Status, financing
Armenia	Metsamor 3	AES-92	
Hungary	Paks II-2	VVER-1200/V-529	Finance provided to cover 80% of cost
Uzbekistan	Farish (Jizzakh)	VVER-1000 + RITM-200N	
India	Kudankulam 7&8	AES-2006	

Country	Plant	Type	Status, financing
Bulgaria	Belene/Kozloduy 7	AES-92	Cancelled
Ukraine	Khmelnitski	completion of 2 x V-392B reactors	Was due to commence construction 2015, 85% financed by loan, but contract rescinded by Ukraine in 2015
South Africa	Thyspunt	up to 8 x AES-2006	Broad agreement signed, no specifics, Russia offers finance, prefers BOO. On hold
Nigeria		AES-2006?	Broad agreement signed, no specifics, Russia offers finance, BOO
Argentina	Atucha 5?	AES-2006	Broad agreement signed, no specifics, Russia offers finance
Indonesia	Serpong	10 MWe HTR	Concept design by OKBM Afrikantov
Algeria	?	?	Agreement signed, no specifics
Jordan	Al Amra	2 x AES-92	Cancelled in 2018
Finland	Hanhikivi	AES-2006	Cancelled
Vietnam	Ninh Thuan 1	4 x AES-2006	Cooperation resumed 2025

AES-91 & AES-92 have 1000 MWe class reactors, AES-2006 has 1200 MWe class reactors.

Russia's policy for building nuclear power plants in non-nuclear weapons states is to deliver on a turnkey basis, including supply of all fuel and repatriation of used fuel for the life of the plant. The fuel is to be reprocessed in Russia and the separated wastes returned to the client country eventually. Evidently India is being treated as a weapons state, since Russia will supply all the enriched fuel for Kudankulam, but India will reprocess it and keep the plutonium.

Considerable export potential for floating nuclear power plants (FNPP), on a fully-serviced basis, has been identified. Indonesia is one possible market. In August 2015 Rosatom and Indonesia's BATAN signed a cooperation agreement on construction of FNPPs.

Since 2006 Rosatom has actively pursued cooperation deals in South Africa, Namibia, Chile and Morocco as well as with Egypt, Algeria, Kuwait, Cambodia, Saudi Arabia, Zambia and Paraguay.

For other fuel cycle exports see companion page on Russia's Nuclear Fuel Cycle.

Aluminium and nuclear power

A number of proposals for aluminium smelters to be co-located with nuclear power plants have been examined, but have not proceeded.

In 2006 the major aluminium producer SUAL (which in March 2007 became part of RUSAL) signed an agreement with Rosatom to support investment in new nuclear capacity at **Kola,** to power expanded aluminium smelting there from 2013. Four units totalling 1000 MWe were envisaged for Kola stage 2 underpinned by a 25-year contract with SUAL, but did not proceed.

From 2007 Rosatom and RUSAL conducted a feasibility study on a nuclear power generation and aluminium smelter at **Primorye in Russia's far east.** The proposal was for four 1000 MWe reactors and a 600,000 t/yr smelter with Atomstroyexport having a controlling share in the nuclear side. The smelter would have required about one third of the output from 4 GWe, and electricity exports to China and North and South Korea are envisaged.

In October 2007 a $8 billion project was announced for the world's biggest aluminium smelter at **Balakovo** in the Saratov region, complete with two new nuclear reactors to power it. The 1.05 million tonne per year aluminium smelter was to be built by RUSAL and would have required about 15 billion kWh/yr. The initial plan was for the existing Balakovo nuclear power plant of four 950 MWe reactors to be expanded with two more, already partly constructed* – the smelter would have required a little over one-third of the output of the expanded power plant. However, in February 2010 it was reported that RUSAL proposed to build its own 2000 MWe nuclear power station, **Balakovo AES2**, with construction to start in 2011. The overall budget for the energy and metals complex was estimated by the Minister of Investment in the Saratov District to be about $12 billion.

* Construction of Phase II of Balakovo plant, started in 1987, was stopped in 1992. At that time, unit 5 was 60% complete and unit 6 was 15% – both VVER-1000. From mid 2000 Rosenergoatom prepared Balakovo II for construction completion. However, then Rusal decided against the plan, and

in 2009 Rosatom announced freezing the project. In 2015 it called for bids to mothball the project by 2019.

Meanwhile, and relevant to these proposals, in 2011 Renova's Integrated Energy Systems (IES) Holding, Russia's largest privately-owned power producer and supplier, agreed to sell its 141 MWe Bogoslovskaya CHP plant to RUSAL in mid-2012, along with the rights to develop a new 230 MWe combined cycle gas turbine unit at the plant, in the central region of Sverdlovsk. This deal, along with another for a supply contract from the Federal Grid Company, enabled RUSAL's Bogoslovosk smelter to continue operating. These arrangements were made at presidential level, and absolved the Bogoslovskaya smelter from paying the cross-subsidy from industrial consumers to other electricity users that is inherent in the general distribution tariff.

Nuclear icebreakers and merchant ship

Nuclear propulsion has proven technically and economically essential in the Russian Arctic where operating conditions are beyond the capability of conventional icebreakers. The power levels required for breaking ice up to 3 metres thick, coupled with refuelling difficulties for other types of vessel, are significant factors. The nuclear fleet has increased Arctic navigation on the Northern Sea Route (NSR) from two to ten months per year, and in the western Arctic, to year-round. In 2020 freight traffic along the NSR reached almost 33 million tonnes and was projected to reach 80 million tonnes by 2024. Greater use of the icebreaker fleet is expected with developments on the Yamal Peninsula and further east. For instance the Yamal LNG project is expected to need 200 shipping movements per year from Sabetta at the mouth of the Ob river.

The fleet is operated by Atomflot, a Rosatom division, and is commercially vital to northern mineral and oil and gas developments, as well as enabling the shortest route from Europe to East Asia. The newest icebreakers being built have a 34-metre beam, able to open a path for large ships.

The icebreaker *Lenin* was the world's first nuclear-powered surface vessel (20,000 dwt) and remained in service for 30 years (1959-89), though new OK-900 reactors (2 x 159 MWt) were fitted in 1970.

It led to a series of larger icebreakers, the six 23,500 dwt *Arktika*-class, launched from 1975. These powerful vessels have two 171

MWt OK-900A reactors delivering 54 MW at the propellers and are used in deep Arctic waters. The *Arktika* was the first surface vessel to reach the North Pole, in 1977. The seventh and largest *Arktika*-class icebreaker – 50 Years of Victory (*50 Let Pobedy*) entered service in 2007. It is 25,800 dwt, 160 m long and 20m wide, and is designed to break through ice up to 2.8 metres thick. Its performance in service has been impressive.

For use in shallow waters such as estuaries and rivers, two shallow-draught *Taymyr*-class icebreakers of 18,260 dwt with one reactor delivering 35 MW were built in Finland and then fitted with their nuclear steam supply system in Russia. They are built to conform with international safety standards for nuclear vessels and were launched from 1989.

Larger third-generation 'universal' LK-60 icebreakers (project 22220) are being built as dual-draught (8.55 or 10.5m) wide-beam (34m) ships of 25,450 dwt or 33,540 dwt with ballast, able to handle 2.8 metres of ice, for use in the Western Arctic year-round and in the eastern Arctic in summer and autumn. In August 2012 the United Shipbuilding Corporation (USC) won the contract for the first new-generation LK-60 icebreaker, *Arktika*. These are powered by two RITM-200 reactors of 175 MWt each, together delivering 60 MW at the propellers via twin turbine-generators and three electric motors. They are being built by USC subsidiary Baltijsky Zavod Shipbuilding in St Petersburg.

In January 2013 Rosatom called for bids to build two more of these universal icebreaker vessels, for delivery in 2019 and 2020, and in May 2014 a contract for RUR 84.4 billion ($2.4 billion) was signed with USC, the vessels to be built at the same Baltic shipyard. In August 2013 Rostechnadzor licensed the shipyard to install the RITM-200 reactor units from OKBM Afrikantov for the pilot model. The keel of *Arktika* was laid in November 2013, and that of *Sibir* in May 2015, *Ural* in July 2016, and *Yakutiya* in May 2020. *Arktika* was launched in June 2016 and *Sibr* in September 2017. *Arktika* entered service in October 2020, *Sibir* in 2021, *Ural* in 2022, and *Yakutiya* in early 2025. A fifth vessel, *Chukotka*, was laid down in December 2020 and launched in November 2024, and the keels of *Leningrad* and *Stalingrad* were laid down in 2024 and 2025 respectively.

A more powerful LK-120 icebreaker (project 10510) delivering 120 MW at four propellers is being designed, capable of breaking through 4.5 metre thick ice, or 2m thick ice at 14 knots. It is for deep-sea use especially in the eastern Arctic and will be 205 m long, 50 m wide and with 13 m draft, of 55,600 dwt. It will be

powered by two RITM-400 reactors of 315 MWt each together delivering 120 MW propulsion. The first vessel will be the *Leader* (or *Lider*). The 50 m beam is to match large tankers. Since they are too big for the St Petersburg shipyard, they are being built at the Zvezda shipyard in the Far East Primorye region, with a target service date of 2030.

LK-60 icebreaker (Rosatom)

The LK-60 is too big for an easy operation around the oil and gas fields, so Project 10570 is under development with LK-40 intended for shallow water and the Arctic shelf. It will displace 20,700 t and be 152 m long, 31 m wide, draft 8.5 m, using a single RITM-200B reactor of 209 MWt delivering 40 MW propulsion.

In 1988 the *NS Sevmorput* was commissioned in Russia, mainly to serve northern Siberian ports. It is a 61,900 tonne 260 m long lash-carrier (taking lighters to ports with shallow water) and container ship with ice-breaking bow. It is powered by the same KLT-40 reactor as used in larger icebreakers, delivering 32.5 propeller MW from the 135 MWt reactor and it needed refuelling only once to 2003. In 2014-15 it was refurbished to give it a service life to at least 2030.

Russian experience with nuclear powered Arctic ships totalled about 360 reactor-years in 2019. In 2008 the Arctic fleet was transferred from the Murmansk Shipping Company under the Ministry of Transport to FSUE Atomflot, under Rosatom.

In February 2023, as part of Russia's Project 22220, an agreement was signed between Baltic Shipyard and Atomflot for the construction of two more nuclear-powered icebreakers. The icebreakers have 60 MW of power (on shafts) and 40 years of expected service life, with commissioning scheduled for December 2028 and December 2030.

Appendix

Earlier plans for new capacity

Rosatom's initial proposal for a rapid expansion of nuclear capacity in the 2000s was based on the cost effectiveness of completing the 9 GWe of then (c. 2002) partially built plant. To get the funds, Minatom offered Gazprom the opportunity to invest in some of the partly completed nuclear plants. The rationale was that the $7.3 billion required for the whole 10 GWe (including the just-completed Rostov 1) would be quickly recouped from gas exports if the new nuclear plant reduced the need to burn that gas domestically.

In September 2006 Rosatom announced a target of nuclear providing 23% of electricity by 2020, thus commissioning two 1200 MWe plants per year from 2011 to 2014 and then three per year until 2020 – adding some 31 GWe and giving some 44 GWe of nuclear capacity in 2020. The Minister of Finance strongly supported the program to increase nuclear share from 15.6% to 18.6% of total in 2020, hence improving energy security as well as promoting exports of nuclear power technology. After 2015 all funding would be from Rosatom revenues.

In September 2007 an ambitious federal target program (FTP) to 2020 was released, working up to over 4 GWe per year new additions from 2016, but noting that from 2012 to 2020 only two 1200 MWe units per year were within the "financial capacity of the federal task program". In February 2008, under the broader Master Plan for Electric Energy Facilities to 2020, the earlier FTP to 2020 was endorsed with little change except that an extra five VVER-1200 units were added as "maximum scenario" or "extra" in the last few years to 2020. As well as the 4800 MWe capacity then under construction, a further 12,000 MWe was planned for completion mostly by 2016, and then a lot more by 2020. Several new sites were involved. Also the new 300 MWe units were listed as being VBER-300 PWR types.

Kursk 5 RBMK was in the FTP to 2009 but construction was halted in 2012, when about 70% complete, and it is mothballed. (see above)

By April 2009 plans were radically scaled back, due both to reduced electricity demand growth and financial constraints. By July

2012, 30.5 GWe nuclear was projected for 2020. This was confirmed in a January 2015 'roadmap', with an average of one reactor per year commissioned to 2025, including the first three TOI units and excluding the Baltic plant. The 'roadmap' excluded smaller and experimental units. But net additions to 2020 were only 6 GWe, taking the target to 31 GWe then.

Russian Nuclear Reactor Planned Additions and Retirements to 2035

Source: Rosatom, January 2015

More significantly, in about 2008 the Ministry of Industry and Energy (MIE) and Rosatom were charged with promptly developing an action plan to attract investment into power generation. It was envisaged that by 2020 much generation would be privatized and competitive, while the state would control natural monopoly functions such as the grid.

In March 2011 the State Duma's energy committee recommended construction of Kursk II with standard VVER-TOI reactors and updating FTP plans to have units 1&2 put online in 2020 and 2023. It said that unit 1 must be in service by the time the first RBMK unit of phase I is closed, to ensure adequate supply to Moscow.

The FTP was based on VVER technology at least to the 2030s. But it highlighted the goal of moving to fast neutron reactors and closed fuel cycle, for which in 2010 Rosatom proposed two options, outlined in the Transition to Fast Reactors section. In stage 1 of the second option, which was adopted, a 100 MWe lead-bismuth-cooled fast reactor was to be built, though this has now been dropped, and in stage 2 over 2015-2020 a pilot demonstration power facility (PDPF) 300 MWe lead-cooled BREST reactor and a multi-purpose fast neutron research reactor (MBIR) are to be built.

In 2009 Siemens announced that it would withdraw from Areva and forge a link with Rosatom. A memorandum of understanding then confirmed the intent to set up a joint venture with Rosatom as majority shareholder, developing Russian VVER designs, building new nuclear power plants, and upgrading existing nuclear plants. This was hailed by Mr Putin as a long-term strategic partnership. However, finalising the agreement was delayed pending Siemens disengaging from Areva, and in September 2011 Siemens announced that it would not proceed. In any case most of Siemens intellectual property remained with Areva, so it would have had little to contribute to Rosatom/Atomenergoprom.

In October 2014 Rosatom resolved in principle to develop small and medium power reactors, though initially they are not expected to compare economically with larger units. In May 2014 Rosenergoatom was completing comparative assessment of VVER-600 from Gidropress and VBER-600 from OKBM designs. In 2016 the VVER-600 was ordered to be built at Kola initially.

In May 2015 the Ministry of Economic Development announced a “very significant" delay in commissioning new nuclear power plants due to “a current energy surplus”. Commissioning of two new Leningrad units and two new Novovoronezh units was delayed by one year, and construction of Smolensk II was postponed for six years. In September 2015 Rosatom said it expected to commission 15 further reactors of 18.6 GWe by 2030, reaching 44 GWe then (so presumably no retirements).

In August 2016 a government decree set out plans to build 11 new reactors beyond Kursk and those then under construction by 2030, as part of the Unified Energy System of Russia. It brought forward the dates for the first two BN-1200 reactors.

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